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United States Patent	12402835
Kind Code	B2
Date of Patent	September 02, 2025
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Catheter and actuator systems and methods

Abstract

The disclosure provides various embodiments of systems to facilitate the cutting of tissue structures and other tissue structures percutaneously.

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Appl. No.: 18/390711

Filed: December 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240206815 A1	Jun. 27, 2024

Related U.S. Application Data

continuation parent-doc WO PCT/US2023/084621 20231218 PENDING child-doc US 18390711
us-provisional-application US 63605856 20231204
us-provisional-application US 63533271 20230817
us-provisional-application US 63496566 20230417
us-provisional-application US 63477317 20221227

Publication Classification

Int. Cl.: A61B18/14 (20060101); **A61B5/00** (20060101); **A61B5/283** (20210101); **A61B18/00** (20060101); A61B17/00 (20060101); A61M25/00 (20060101)

U.S. Cl.:

CPC **A61B5/6859** (20130101); **A61B5/283** (20210101); **A61B18/1492** (20130101); A61B2017/0003 (20130101); A61B2018/00279 (20130101); A61B2018/00369 (20130101); A61B2018/00601 (20130101); A61B2018/00839 (20130101); A61M25/0074 (20130101)

Field of Classification Search

CPC: A61B (5/283); A61B (5/6859); A61B (2018/00225); A61B (2018/0038); A61B (2017/00225); A61B (2017/0003)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present patent application claims the benefit of priority to and is a continuation of International Patent Application No. PCT/US2023/84621, filed Dec. 18, 2023, which in turn claims the benefit of priority to U.S. Provisional Patent Application No. 63/477,317, filed Dec. 27, 2022, U.S. Provisional Patent Application No. 63/496,566, filed Apr. 17, 2023, U.S. Provisional Patent Application No. 63/533,271, filed Aug. 17, 2023, and U.S. Provisional Patent Application No. 63/605,856, filed Dec. 4, 2023. Each of the aforementioned patent applications is incorporated by reference herein in its entirety for all purposes.

BACKGROUND

(1) The disclosure relates generally to medical treatment devices and techniques, and, in some aspects, to methods and devices for diagnosis and treatment of myocardial tissue. The present disclosure provides improvements over the state of the art.

SUMMARY OF THE DISCLOSURE

(2) BASILICA and LAMPOON are aortic and mitral leaflet laceration procedures that use transcatheter electrosurgery. A guidewire traverses potentially obstructive heart valve leaflet tissue and then the inner-curvature of the kinked guidewire traversing the leaflet is electrified during traction to accomplish a longitudinal split of the leaflet.

(3) Left ventricular outflow tract (LVOT) obstruction complicates hypertrophic cardiomyopathy and transcatheter mitral valve replacement. Septal reduction therapies including surgical myectomy and alcohol septal ablation are limited by surgical morbidity or coronary anatomy and high pacemaker rates respectively. Applicants have developed a novel transcatheter procedure, mimicking surgical myotomy, called SESAME (SEptal Scoring Along the Midline Endocardium). The SESAME procedure uses an insulation-modified guidewire to lacerate myocardium (heart muscle) instead of heart leaflet tissue, using a different system design from the BASILICA and LAMPOON procedures. In some aspects, the SESAME electrosurgical procedure can include an asymmetric insulation gap astride the guidewire kink, or bend. The kink or bend concentrates electrical charge and helps to position the charge-delivery-device at the therapy target to avoid bystander injury. The insulation gap, discussed below; is intended to overcome the tendency of charge to concentrate on the outer aspect of a kink.

(4) In accordance with the present disclosure, implementations of a device to cut tissue are provided. Some such implementations include an elongate body having a proximal end and a distal end, an elongate tether operably coupled to the elongate body, wherein the elongate tether and elongate body are configured to be longitudinally displaceable with respect to one another. The device can further include a cutting element disposed on at least one of the elongate body and elongate tether, wherein relative longitudinal movement of the elongate tether and the elongate body causes the cutting element to cut through anatomical tissue that the device is placed adjacent to.

(5) In some implementations, the elongate tether can be configured to be received at least partially within a lumen defined in the elongate body. The cutting element can include at least one blade that is configured to cut through tissue. The cutting element can include an electrically conductive element configured to be coupled to an electrical power source in order to electrify the electrically conductive element.

(6) The cutting element can be disposed on an outer surface of the elongate body. Adjacent anatomical tissue can be cut by the cutting element as the elongate body is advanced or retracted along the elongate tether. In some implementations, the cutting element can include at least one blade that is configured to cut through tissue. The cutting element can include an electrically conductive element configured to be coupled to an electrical power source in order to electrify the electrically conductive element. The electrically conductive element can include at least one supply electrode configured to physically contact tissue to be cut. The supply electrode can be configured

to be operably coupled to an electrical power supply to supply current to the supply electrode.

(7) In some implementations, the device can further include a return electrode configured to direct current from a region proximate the supply electrode back to the electrical power supply. The return electrode is operably coupled to the tether. In some implementations, the device can further include a tissue anchor disposed at a distal end of the elongate tether. If desired, the return electrode is operably coupled to the anchor. The tether can include a return conductor along its length operably coupled to the return electrode. The return electrode can include the anchor. The at least one supply electrode can be disposed on an outer circumferential surface of the elongate body. The at least one supply electrode can be disposed on a distal tip of the elongate body. The at least one supply electrode can at least partially surround the tether.

(8) In some implementations, the tether can exit through a distal end of the elongate body. In some implementations, the tether can exit through a lateral side port defined through a side wall of the elongate body proximate a distal end region of the elongate body. The device can further include a plurality of markers disposed along the tether separated by a predetermined spacing. In some implementations, the device can further include a guidewire passage disposed along at least a portion of a length of the elongate body. The guidewire passage can extend to a distal tip of the elongate body. The tether can exit the elongate body through a lateral side port defined in the elongate body. The tissue anchor and elongate tether can be coupled to a proximal anchor. The elongate tether can be movably disposed within a lumen of the elongate body along a majority of the length of the elongate body. The elongate body can be slidably disposed within a lumen of at least one outer deflatable catheter.

(9) In some implementations, the device can further include an actuator assembly operably coupled to a proximal portion of the elongate body and to a proximal region of the elongate tether. The actuator assembly can be configured to permit a user to selectively move the elongate body with respect to the elongate tether. In some implementations, the actuator assembly can include (i) a first actuator operably coupled to a proximal end of the elongate body and (ii) a proximal anchor to secure a proximal region of the elongate tether. The first actuator can be configured to longitudinally displace the elongate member with respect to the proximal anchor to permit the elongate body to be selectively moved proximally and distally with respect to the proximal anchor. The elongate body can be configured to ride along the elongate tether.

(10) In some implementations, the actuator assembly can further include a second actuator operably coupled to a proximal end of a first outer catheter. The first outer catheter can define a lumen along its length that surrounds the elongate member and the tether. The second actuator can be actuated to longitudinally displace the first outer catheter with respect to the elongate tether and the elongate body. The actuator assembly can further include a third actuator operably coupled to a proximal end of a second outer catheter. The second outer catheter can define a lumen along its length that surrounds the first outer catheter, the elongate member and the tether. The third actuator can be actuated to longitudinally displace the second outer catheter with respect to the elongate tether, the elongate body and the first outer catheter.

(11) In some implementations, the first outer catheter and the second outer catheter can include active or passive steering mechanisms that permit the distal end region of each said catheter to be actively steered by a user. The elongate member can include an inner catheter that includes a steering mechanism to permit the distal end region of the inner catheter to be actively steered by a user. In some implementations, the elongate tether can be configured to be coupled to a distal anchor configured to be deployed into tissue proximate a distal end of the device. The proximal anchor can include a tensioner to selectively apply tension to the tether when the distal anchor is deployed into the tissue. The inner catheter can be configured to slide proximally and distally over the tether after the tether is tensioned. The first outer catheter can be configured to slide proximally and distally over the inner catheter. The second outer catheter can be configured to slide proximally and distally over the first outer catheter. Each of the proximal anchor, first actuator, second actuator

and third actuator can be operably coupled to a respective carrier. Each respective carrier can be configured to slide on a common guide rail.

(12) In some implementations, a proximal end of the inner catheter can be configured to be lifted out of the first actuator. A proximal end of the first outer catheter can be configured to be lifted out of the second actuator. A proximal end of the second outer catheter can be configured to be lifted out of the third actuator. The tensioner can include a first body coupled to the common guide rail and a second body that is movable with respect to the first body, wherein the second body is fixedly coupled to the tether. The second body can be coupled to the first body by an elastic member, such as a tension spring or a coiled spring, such as a flat spring. The second body can be movable from a first position wherein the tether is not tensioned to a second position wherein the tether is tensioned. The second body can be moved from the first position to the second position along a linear path. The second body can be moved from the first position to the second position along a curved path.

(13) In some implementations, the first outer catheter can include a distally deployable anchor wire to facilitate anchoring a distal end of the first outer catheter in a desired location. The tether can include a tubular member that defines a passageway therethrough. The tissue anchor can be operably coupled to the distal end of the tether by way of a flexible coupling. The flexible coupling permits the tissue anchor to swivel with respect to the distal end of the tether. The at least one supply electrode can include at least one electrode configured to supply current, at least one electrode to return current to a power source to complete a circuit, and at least one sensing electrode; which can comprise the same physical electrodes, or different physical electrodes.

(14) In some implementations, a radiopaque marker can be provided proximate the lateral side port to permit a user to determine the longitudinal and rotational orientation of the lateral side port. The elongate tether can include an elongate tubular member, and the tissue anchor can be configured to be deployed out of a distal port of the elongate tubular member. The tissue anchor can be pivotally coupled about a pivot point to a distal end of an elongate inner member disposed within the elongate tubular member. The device can further include a tension member coupled proximate a distal end of the tissue anchor. The tension member can be directed proximally through the elongate tubular member to be externalized from a patient. Applying tension to the tension member causes the anchor to articulate about the pivot point until at least one tine of the tissue anchor points along a proximal direction. The device can further include at least one visualization marker proximate a distal end or an exit port of at least one of the elongate body, the first outer catheter, the second outer catheter, the elongate tether, and the anchor. One or more of the visualization markers can include radiopaque material. One or more of the visualization markers can be configured to be visible under a magnetic resonance imaging modality. In some implementations, one or more of the first actuator, second actuator and third actuator can be operably coupled to an automated surgical device.

(15) In some implementations, the disclosure provides implementations of a medical device including an elongate body having a proximal end and a distal end, and an elongate tether operably coupled to the elongate body. The elongate tether and elongate body can be configured to be longitudinally displaceable with respect to one another. The device can further include an electrode disposed on at least one of the elongate body and elongate tether, and electrical circuitry operably coupled to the electrode. The electrical circuitry can be configured to determine a state of at least one of the medical device and the anatomical tissue. The electrical circuitry can be configured to detect an incoming signal from the anatomical tissue to confirm that the electrode is in physical contact with the anatomical tissue. The incoming signal from the anatomical tissue can include an electrocardiogram signal from cardiac tissue. If desired, the electrical circuitry can be configured to detect a voltage or current drop across the electrode after electrical power has been applied to the electrode. In some implementations, relative longitudinal movement of the elongate tether and the elongate body can cause the electrode to cut through anatomical tissue that the device is placed

adjacent to when the electrode is energized. The electrical circuitry can be configured to correlate the voltage or current drop with a state selected from the group consisting of (i) a state of tissue being cut by the electrode, (ii) a state of fouling of the electrode. The device can further include a pressure sensor located proximate the electrode. The pressure sensor can be operably coupled to the processor. The processor can be programmed to determine at least one biological parameter based on receiving a signal from the pressure sensor, the at least one biological parameter.

(16) The elongate body can be configured to be held stationary adjacent anatomical tissue. The at least one cutting element can be disposed on the elongate tether. The elongate tether and cutting element can be configured to be slid alongside or within the elongate body in a reciprocating manner while the elongate body is held in a stationary position adjacent the anatomical tissue. The elongate tether can be configured to be received at least partially within a lumen defined in the elongate body. The elongate tether can be configured to exit from a proximal exit port formed in the elongate body and further wherein the elongate tether can be configured to re-enter the elongate body in a distal entrance port. The elongate body can be configured to be bent into a deployed configuration along a region that includes the proximal exit port and the distal entrance port. The elongate body can be bent into the deployed configuration, and the elongate tether can be directed away from the elongate body when the elongate tether is under tension.

(17) In some implementations, a distal end of the elongate tether is coupled to a distal portion of the elongate body by an elastic element that can stretch longitudinally, and further wherein the elastic element is configured to retract the elongate tether in a distal direction when tension is reduced on the proximal end of the tether. The elastic element can include a tension spring. The elongate body can be configured to be deformed at least partially around an anatomic structure to be cut by the device. The elongate tether can include a cutting electrode mounted thereon configured to be coupled to an electrosurgical power source. In some implementations, the device can further include a depth sensing electrode to sense the depth of the tissue that the elongate tether is passing through. In some implementations, the elongate body can include a flexible distal section disposed distally of the distal entrance port to conform to the anatomy of a patient's ventricle. If desired, the elongate tether can pass around a bearing surface located on or within the elongate body located distally of the distal entrance port, and the elongate tether can pass through the elongate body to permit both ends of the elongate tether to be externalized from a patient while performing a cutting operation in the patient's heart. The elongate body can define a first elongate body and a proximal end of the elongate tether can be coupled to a second elongate body configured to move alongside or at least partially within the first elongate body. The elongate tether can include at least one of a radiopaque wire, a radiopaque suture material, a textured body, a radiofrequency (RF) electrode and a razor wire. In some implementations, the elongate tether can be operably coupled to an outer tubular member that is configured to be advanced proximally and distally over the elongate body. The elongate tether can be configured to cause the outer tubular member to form a bowed shape when tension is applied to the elongate tether. The elongate tether can be configured to be biased laterally away from the outer tubular member when the outer tubular member is formed into the bowed shape to permit the at least one cutting element disposed on the elongate tether to cut through tissue as the outer tubular member is advanced proximally and distally over the elongate body. In some implementations, the elongate body can define a tissue anchor that deploys along a proximal direction from a first location wherein the tissue anchor is disposed at least partially within the elongate body to a second location wherein the tissue anchor is advanced outwardly from the elongate body along a proximal direction into a tissue mass.

(18) In further accordance with the disclosure, implementations of a medical device are provided. The medical device can include an elongate tether having a proximal end and a distal end. The medical device can include a first tubular catheter surrounding the elongate tether and be slidably displaceable along and with respect to the elongate tether. The first tubular catheter can have a proximal end and a distal end. The medical device can include a second tubular catheter

surrounding the first tubular catheter and be slidably displaceable along and with respect to the first tubular catheter. The second tubular catheter can have a proximal end and a distal end. The medical device can include a third tubular catheter surrounding the second tubular catheter and be slidably displaceable along and with respect to the second tubular catheter. The third tubular catheter having a proximal end and a distal end.

(19) In further accordance with the disclosure, the medical device can further include an actuator assembly coupled to a respective proximal end of one or more of the elongate tether, the first tubular catheter, the second tubular catheter, and the third tubular catheter, where provided. In some implementations, the medical device can further include a tissue anchor operably coupled to the distal end of the elongate tether. The actuator assembly can include a proximal anchor operably coupled to the proximal end of the elongate tether. The proximal anchor can be configured to permit tension to be applied to the elongate tether when the tissue anchor is anchored in tissue.

(20) In further accordance with the disclosure, the actuator assembly can further include a first actuator operably coupled to a proximal end of the first tubular catheter, wherein the first actuator is configured to advance and retract the first tubular catheter proximally and distally over the elongate tether. The actuator assembly can further include a second actuator operably coupled to a proximal end of the second tubular catheter. The second actuator can be configured to advance and retract the second tubular catheter proximally and distally over the first tubular catheter. The distal end of the second tubular catheter can be capable of being retracted proximally past the distal end of the first tubular catheter. The actuator assembly can further include a third actuator operably coupled to a proximal end of the third tubular catheter. The third actuator can be configured to advance and retract the third tubular catheter proximally and distally over the second tubular catheter. The distal end of the third tubular catheter can be capable of being retracted proximally past the distal end of the first tubular catheter and the distal end of the second tubular catheter.

(21) In some implementations, at least one of the first tubular catheter, the second tubular catheter, and the third tubular catheter can include active or passive steering mechanisms that permit the distal end region of each said catheter to be actively steered by a user. Each of the proximal anchor, first actuator, second actuator and third actuator can be operably coupled to a respective carrier in the actuator assembly, and each respective carrier can be configured to slide on a common guide rail. The relative longitudinal position of each of the proximal anchor, first actuator, second actuator and third actuator can be longitudinally adjusted along the common guide rail with respect to other components mounted on the common guide rail. The device can further include at least one cutting element disposed on at least one of the elongate tether, first tubular catheter, second tubular catheter, and third tubular catheter. Relative longitudinal movement of the at least cutting element with respect to another component of the device can cause the cutting element to cut through anatomical tissue that the cutting element is placed adjacent to.

(22) In some implementations, the device can further include at least one electrode disposed on at least one of the elongate tether, first tubular catheter, second tubular catheter, and third tubular catheter, wherein relative longitudinal movement of the at least one electrode with respect to another component of the device causes the at least one electrode to cut through anatomical tissue that the at least one electrode is placed adjacent to. The device can further include at least one visualization marker disposed on at least one of the elongate tether, first tubular catheter, second tubular catheter, and third tubular catheter. The at least one visualization marker can be used to visualize a distal region of the device while inside a patient to permit a user of the device to determine the relative axial and rotational location of different component systems relative to each other and surrounding anatomy.

(23) The disclosure further provides implementations of method of performing a medical procedure. The method can include providing a device as described herein, directing the device according to a target location inside a patient, placing the distal end of the third tubular catheter in a first location, placing the distal end of the second tubular catheter in a second location located

distally with respect to the first location, placing the distal end of the first tubular catheter in a third location located distally with respect to the second location, and performing a therapeutic or diagnostic procedure using at least one of the first tubular catheter, the second tubular catheter and the third tubular catheter.

(24) In some implementations, a distal region of the third tubular catheter can be disposed in an aortic arch of a patient, a distal region of the second tubular catheter can be disposed through a cardiac valve of a patient, and the first tubular catheter can be manipulated to perform a therapeutic or diagnostic procedure. The therapeutic procedure can include cutting into a left ventricular outflow tract obstruction to increase the effective cross-sectional area of the left ventricular outflow tract. In some implementations, the first tubular catheter can include a microcatheter, and the method can include directing the first tubular catheter into the right ventricle of the patient, through the septum of the patient into the left ventricle of the patient, and cutting into a left ventricular outflow tract obstruction to increase the effective cross-sectional area of the left ventricular outflow tract. In some implementations, the first tubular catheter can include a delivery catheter to deliver a beneficial agent or a medical device to an anatomical location. The first tubular catheter can include a cutting element to cut through anatomical tissue. The diagnostic or therapeutic procedure can be selected from the group including a MIRTH procedure, a LAMPOON procedure, an ANTEPASTA procedure, an ELASTIC procedure, a robotic surgical procedure, a cerclage procedure, or delivery of a medical device. The medical device can be selected from the group consisting of a stent and an artificial valve, such as an artificial cardiac valve including but not limited to an artificial mitral, aortic, tricuspid, or pulmonary valve, for example. The device is introduced into the patient by way of a femoral access point, a jugular access point, a carotid access point, or an apical access point through a wall of the heart, for example.

(25) In some implementations, the at least one supply electrode can include a fin shaped electrode located on a dorsal surface proximate a distal end of the elongate body. If desired, the device can further include a flush port configured to flow a flushing fluid over the electrode when the at least one supply electrode is electrified. The device can further include a steering wire configured to cause a distal region of the device to laterally deflect.

(26) The foregoing and other features and advantages of the disclosed technology will become more apparent from the following detailed description of several embodiments which proceeds with reference to the accompanying figures.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 is a representative illustration of a three-dimensional cardiac computed tomography image with a virtually embedded artificial SAPIEN™ valve in the mitral position to guide in predicting LVOT obstruction.

(2) FIGS. 2A-2B are representative echocardiograms illustrating the presence of a flow obstruction within the left ventricular outflow tract that contributes to blocking blood flow to the aorta of a patient.

(3) FIG. 3 is a representative computerized tomography (CT) scan illustrating a cross section of a patient's heart indicating the location of left ventricular outflow tract obstruction (LVOTO).

(4) FIG. 4 is a further a representative computerized tomography (CT) scan illustrating a cross section of a patient's heart indicating a representative path of catheters to access the left ventricle of a patient.

(5) FIG. 5 is a representative implementation of a catheter-based system to perform a percutaneous procedure to cut tissue in accordance with the present disclosure using a reciprocating cutter.

(6) FIG. 6 illustrates aspects of the catheter-based system of FIG. 5 in accordance with some

aspects of the present disclosure.

(7) FIGS. 7A-7C illustrate still further aspects of the catheter-based system of FIG. 5 in accordance with some aspects of the present disclosure.

(8) FIGS. 8A-8F illustrate features of the catheter-based system of FIG. 5 to confirm device orientation while performing a procedure in accordance with the present disclosure.

(9) FIGS. 8G-8P depict aspects of a further implementation of a distal tip for the catheter-based system in accordance with the present disclosure, or aspects thereof.

(10) FIGS. 9A-12C illustrate aspects of a representative procedure to treat a LVOTO in accordance with some aspects of the present disclosure.

(11) FIGS. 13A-13C illustrate results of an illustrative LVOTO treatment procedure in a porcine model in accordance with some aspects of the present disclosure.

(12) FIG. 14 is an illustrative implementation of an anchor in accordance with the present disclosure.

(13) FIG. 15 is an illustration of a further system in accordance with the present disclosure for cutting tissue using a reciprocating cutter.

(14) FIG. 16 is a schematic of anatomy of a heart illustrating a target cutting site for treating a LVOTO in accordance with the present disclosure.

(15) FIGS. 17A-17B are schematics illustrating aspects of introducing a system into a patient's heart to treat a LVOTO in accordance with the present disclosure.

(16) FIGS. 18A-18B are schematics illustrating further aspects of a technique to treat a LVOTO in accordance with the present disclosure including delivering an anchor to hold a treatment system in place within a patient's heart.

(17) FIGS. 19A-19C are schematics illustrating further aspects of a technique to treat a LVOTO in accordance with the present disclosure including performing a cutting operation on the LVOTO using a reciprocating cutter.

(18) FIGS. 20A-20B are schematics illustrating further aspects of a technique to treat a LVOTO in accordance with the present disclosure including re-sheathing the cutter and retrieving the anchor of the system.

(19) FIGS. 21A-21B are schematics illustrating further aspects of a technique to treat a LVOTO in accordance with the present disclosure including withdrawing the catheter-based system from the patient and illustrating the treated LVOTO.

(20) FIGS. 22A-22C illustrate aspects of a further technique to treat a LVOTO in accordance with the present disclosure that includes deploying an anchor proximate a distal location of the LVOTO structure.

(21) FIGS. 23A-23C illustrate aspects of yet a further method and system to cut tissue in accordance with the present disclosure utilizing a deflectable cutting catheter.

(22) FIGS. 24A-25C illustrate aspects of still a further technique to treat a LVOTO in accordance with the present disclosure that includes deploying a further implementation of an anchor.

(23) FIGS. 26-30B illustrate aspects of a further method and system to cut tissue in accordance with the present disclosure.

(24) FIG. 31 is a side plan view of a further implementation of an actuator assembly for a catheter system in accordance with the present disclosure.

(25) FIG. 32 is an enlarged upper isometric view of a portion of the actuator assembly of FIG. 31.

(26) FIG. 33 is a further enlarged upper isometric view of a portion of the actuator assembly of FIG. 31.

(27) FIG. 34 is a side view of a handle portion of an actuator assembly of FIG. 31 with a lateral half of the outer housing removed.

(28) FIG. 35 is a view of internal components of an illustrative handle assembly in accordance with the present disclosure.

(29) FIG. 36 is an isometric view of a linear tensioner in accordance with the present disclosure

depicting half of the housing portion removed.

(30) FIG. 37 is an isometric view of the housing component omitted from FIG. 36.

(31) FIG. 38 is an isometric view of a variation of an actuator assembly in accordance with the present disclosure.

(32) FIGS. 39A-39B are different isometric views of the actuator assembly of FIG. 38 mounted to a stool.

(33) FIG. 39C is an isometric view of an actuator assembly in accordance with the present disclosure including two actuators.

(34) FIGS. 40A-40D are different isometric views of a further actuator assembly in accordance with the present disclosure.

(35) FIG. 41 is a side view of a distal portion of a further catheter in accordance with the present disclosure configured to deploy a tissue anchor out through a lateral side port of the catheter.

(36) FIGS. 42A-42D are images of the catheter of FIG. 41 illustrating successive stages of deploying the tissue anchor out through the lateral side port of the catheter.

(37) FIGS. 43-44 are views of the tissue anchor removed from the catheter of FIG. 41.

(38) FIGS. 45A-45B illustrate aspects of utilizing a catheter-based system in accordance with the present disclosure to perform a LAMPOON procedure.

(39) FIGS. 46A-46B illustrate aspects of utilizing a catheter-based system in accordance with the present disclosure to perform a BASILICA procedure.

(40) FIGS. 47A-47B illustrate aspects of utilizing a catheter-based system in accordance with the present disclosure to perform an ELASTIC procedure.

DETAILED DESCRIPTION

(41) The present application presents advantages and improvements over systems described in International Patent Application No. PCT/US2023/060223, filed Jan. 6, 2023. The aforementioned patent application is incorporated by reference herein in its entirety for all purposes.

(42) Implementations in accordance with the present disclosure provide improved systems and methods for cutting tissue, including but not limited to myocardial tissue. Implementations in accordance with the present disclosure can form a lengthwise cut through tissue utilizing a reciprocating cutter that cuts from an outer surface of an obstruction, such as a LVOTO, downwardly into the tissue. This can be contrasted with the techniques described in PCT/US2023/060223, which teaches forming a passageway through tissue near a bottom region of tissue to be cut, and then cutting the tissue above the passageway to complete the cut. It will be appreciated by those of skill in the art that the disclosed implementations are fundamentally different from those described in PCT/US2023/060223 in a variety of ways.

(43) FIGS. 1-4 depict aspects of a left ventricular outflow tract obstruction (“LVOTO” herein). While the cutting of a LVOTO is specifically illustrated, it will be appreciated that the disclosed embodiments can be used for other purposes, including the cutting of myocardial tissue to debulk other cardiac structures, or to perform different percutaneous procedures in the cardiovascular system, a patient's sinus passages, within a patient's neurovascular structures, urinary structures, abdominal structures or digestive structures. Moreover, implementations in accordance with the present disclosure can be utilized in laparoscopic, thoracic, and other procedures.

(44) FIG. 1 is a representative illustration of a three-dimensional cardiac computed tomography image with a virtually embedded artificial SAPIEN™ valve in the mitral position to guide in predicting LVOT obstruction. In some implementations of procedures according to the present disclosure, pre-procedure planning can include generating a dedicated cardiac computed tomography (“CT”) image to measure one or more of a variety of variables to assist in performing a tissue cutting procedure. For example, the CT image can be used to determine, or closely estimate valve size. This can also be facilitated with a transesophageal echocardiogram (“TEE”). It is also possible to measure the actual internal dimension of the valve under visualization which can then be correlated with the true internal dimension of the failed bioprosthetic valve. Comparing the

measured value with the known value of the device can provide a proper basis for scaling the image to more closely measure the actual dimensions of internal cardiac structures. With continuing reference to FIG. 1, it is further possible to measure the aorto-mitral angle, wherein a favorable angle is in excess of 105 degrees. Moreover, it is possible to measure, or to at least estimate, the dimensions of a neo-left ventricular outflow tract (LVOT) area after a “virtual” implantation of a SAPIEN valve, as depicted in FIG. 1. Preferably, the LVOT area will exceed 200 mm.^{sup.2} to prevent LVOT obstruction. If the patient's septum is thick, alcohol septal ablation can be performed ahead of time to debulk the septum somewhat and decrease the risk of LVOT obstruction in cases where the predicted neo-LVOT area is less than 200 mm.^{sup.2}. It is also possible to perform balloon assisted translocation of the mitral anterior leaflet to prevent LVOT obstruction, which is a significant concern of transcatheter mitral valve replacement (“TMVR”). Success rates for TMVR procedures is about 94-97%, with 91-95% 30-day survival and 86% 1-year survival rates.

(45) FIGS. 2A-2B are representative echocardiograms illustrating the presence of a flow obstruction within the left ventricular outflow tract that contributes to blocking blood flow to the aorta of a patient. In FIG. 2A, an LVOT obstruction is blocking blood flow (area inside dashed lines in FIG. 2B) to the aorta by way of the LVOT.

(46) FIG. 3 is a representative computerized tomography (CT) scan illustrating a cross section of a patient's heart indicating the location of left ventricular outflow tract obstruction (“LVOTO”). The procedure described in PCT/US2023/060223 can be performed, wherein an electrified guidewire is navigated through the obstruction by tunneling through the mass starting at a proximal end of the mass (facing away from the ventricular apex) along a direction toward the ventricular apex where the guidewire exits the mass and re-enters the volume of the ventricle, effectively forming a tunnel through a bottom portion of the tissue mass. The distal end of the guidewire can be snared once exiting the tissue mass. The initial trajectory for the guidewire is perpendicular to the surface of the septum to penetrate the sometimes tough, basal septum. As seen in FIG. 3, the pathway of the guidewire is not straight, but curves off the initial vector to avoid crossing into the right ventricle and creating a ventricular septal defect (“VSD”). In FIG. 3, for purposes of clarity, “LA” refers to the left atrium and “LV” refers to the left ventricle.

(47) FIG. 4 is a further representative “three-dimensional” computerized tomography (CT) scan illustrating a cross section of a patient's heart indicating a representative path of catheters to access the left ventricle of a patient. In FIG. 4, the path to the basal end of the intended cut typically starts just to the right side of the left-right commissure of the aortic valve. In view of the foregoing, and the teachings of PCT/US2023/060223, it will be appreciated that further approaches for performing similar procedures allowing additional device control during the procedure can be desirable.

(48) Thus, in further accordance with the present disclosure, a first representative implementation for a system **100** for cutting tissue is depicted in FIG. 5. Further aspects of system **100** are depicted in FIGS. 6-12C.

(49) FIG. 5 is a representative implementation of a catheter-based system **100** to perform a percutaneous procedure to cut tissue in accordance with the present disclosure. While the system **100** utilizes a reciprocating cutter (described below), the system **100** can be configured with different end effectors to perform different procedures. The system **100**, as depicted includes three concentrically disposed deflectable catheters including a radially outermost catheter **110**, within which a second deflectable intermediate catheter **120** is slidably received. A third, innermost deflectable catheter **130** is slidably disposed within deflectable catheter **120**. A tether **140**, or other instrumentality, such as a guidewire, a snare or the like is slidably received within catheter **130**. Each deflectable catheter **110**, **120**, **130** can include one or more steering wires (not shown) to which tension can be applied to selectively cause the distal end (**114**, **124**, **134**) of each respective catheter (**110**, **120**, **130**) to bend in a preferential direction. Each catheter **110**, **120**, **130** is comprised of a tubular body having a proximal end coupled to an actuator and a free distal end.

Each catheter **110, 120, 130** can be moved axially and rotationally with respect to the other components of system **100**. Each catheter **110, 120, 130** is held in relative position by a respective carriage **210, 220, 230** that is slidably received on a rail **201** of a stand **200** of system **100**. Each carriage **210, 220, 230** can be axially translated between the proximal end **202** and the distal end **204** of the rail **201**. In some implementations, one or more of the carriages **210, 220, 230** can be locked in place with respect to the rail **201**. As depicted, each catheter **110, 120, 130** is slidably received along a vertical direction into an upwardly extending portion of a respective carriage in a fork shaped coupling (e.g., **214**). For example, a channel **115** is formed into catheter **100** that is slidably received within fork **214** to prevent axial movement of catheter **210** with respect to carriage **210**. Each carriage (e.g., **210**) is defined by a main body portion (e.g., **212**) that defines a channel (e.g., **218**) therethrough to at least partially surround rail **201** of stand **200**.

(50) Each of catheters **110, 120, 130** can be displaced rotationally about a central axis of system **100** (e.g., about tether **140**) with respect to tether **140**, and each other. It will be appreciated that any of catheters **110, 120, 130** can be utilized with stand **200** alone, or in combination with other system components. Thus, a triple catheter assembly may be used as depicted in FIG. 5, or a single or double catheter assembly may be used if all three catheters are not required. Catheters **110, 120, 130** can be of any desired length. In accordance with some implementations, catheter **110** can be between about 80 cm and about 120 cm in length, or any increment therebetween of about one centimeter. In accordance with further aspects, catheter **120** can be between about 90 cm and about 140 cm in length, or any increment therebetween of about one centimeter. In accordance with still further aspects, catheter **130** can be between about 100 cm and about 160 cm in length, or any increment therebetween of about one centimeter. Tether **140** can be any desired length, such as between about 120 and 300 cm in length, or any increment therebetween of about one centimeter.

(51) With continuing reference to FIG. 5, outermost catheter **110** is comprised of an elongate tubular body **113** having a proximal end **112** operably coupled to an actuator **116** and a free distal end **114** that may be steered by tensioning a steering wire, for example, that is actuated by one or more buttons or levers within the actuator **116**. Outermost catheter **110** further defines a lumen **118** (not shown) along at least a part of its length to slidably receive tubular shaft **123** of catheter **120**. Catheter **110** can be slid along rail **201** and locked in place on rail **201**, if desired. In use, the outer surface of tubular portion **113** of catheter **110** is fluidly sealingly received through an entrance port (not shown) that is fluidly coupled to a patient's anatomy. Consequently, in use, sliding movement of carriage **210** with respect to track or rail **201** results in proximal-distal movement of catheter **110** within a patient's vasculature independently of movement of catheters **120, 130**, or tether **140**.

(52) Intermediate catheter **120**, in turn, is comprised of an elongate tubular body **123** having a proximal end **122** operably coupled to an actuator **126** and a free distal end **124** that may be steered by tensioning a steering wire, for example, that is actuated by one or more buttons or levers within the actuator **126**. Intermediate catheter **120** further defines a lumen **128** (not shown) along at least a part of its length to slidably receive tubular shaft **133** of catheter **130** therein. Intermediate catheter **120** can be slid along rail **201** and locked in place on rail **201**, if desired, as with catheter **110**. In use, the outer surface of tubular portion **123** of catheter **120** is fluidly sealingly received through an entrance port (not shown) located proximally or within actuator/handle **116** of catheter **110** to prevent undesired leakage of fluid between any annular clearance formed between the inner surface of lumen **118** and the outer surface of tubular member **123**. As with catheter **110**, in use, sliding movement of carriage **220** with respect to track or rail **201** results in proximal-distal movement of catheter **120** within a patient's vasculature independently of movement of catheters **110, 130**, or tether **140**. Further, catheter **120** can be displaced rotationally about central axis of system **100** (e.g., about tether **140**) with respect to tether **140**, catheter **110**, and catheter **130**.

(53) With continuing reference to FIG. 5, innermost catheter **130** is comprised of an elongate tubular body **133** having a proximal end **132** operably coupled to an actuator **136** and a free distal end **134** that may be steered by tensioning a steering wire, for example, that is actuated by one or

more buttons or levers within the actuator **136**. Intermediate catheter **130** further defines a lumen **138** (not shown) along at least a part of its length to slidably receive tether **140** therethrough. Innermost catheter **130** can be slid along rail **201** and locked in place on rail **201**, if desired, as with catheters **110**, **120**. In use, the outer surface of tubular portion **133** of catheter **130** is fluidly sealingly received through an entrance port (not shown) located proximally or within actuator/handle **126** of catheter **120** to prevent undesired leakage of fluid between any annular clearance formed between the inner surface of lumen **128** and the outer surface of tubular member **133**. As with catheters **110**, **120**, in use, sliding movement of carriage **230**) with respect to track or rail **201** results in proximal-distal movement of catheter **130** within a patient's vasculature independently of movement of catheters **110**, **120**, or tether **140**. Further, catheter **130** can be displaced rotationally about central axis of system **100** (e.g., about tether **140**) with respect to tether **140**, catheter **110**, and catheter **120**.

(54) Each of the catheters set forth herein (e.g., **110**, **120**, **130**) can be made from a variety of materials, including multilayer polymeric extrusions, such as those described in U.S. Pat. No. 6,464,683 to Samuelson or U.S. Pat. No. 5,538,510 to Fontirroche, the disclosure of each being incorporated by reference herein in its entirety for all purposes. Other structures are also possible, including single or multilayer tubes reinforced by braiding, such as metallic braiding material. Any of the catheters or guidewires disclosed herein or portions thereof can be provided with regions of varying or stepped-down stiffness with length using any of the techniques set forth in U.S. Pat. No. 7,785,318, which is incorporated by reference herein in its entirety for any purpose whatsoever. The catheters herein (e.g., **110**, **120**, **130**) can be provided with these and other structures to enhance pushability and torqueability. The catheters disclosed herein (e.g., **110**, **120**, **130**) can have a varied stiffness along their length, particularly in their distal regions by adjusting the cross-sectional dimensions of the material to impact stiffness and flexibility, while maintaining pushability, as well as the durometer of the material. Hardness/stiffness is described herein with reference to Shore hardness durometer ("D") values. Shore hardness is measured with an apparatus known as a Durometer and consequently is also known as "Durometer hardness". The hardness value is determined by the penetration of the Durometer indenter foot into the sample. The ASTM test method designation is ASTM D2240 00. For example, in some implementations, a more proximal region of the catheter can have a durometer of about 72 D, an intermediate portion of the catheter (the proximal most 20-30 cm of the last 35 cm, for example that typically traverses an aortic arch) can have a durometer of about 55 D, and the distal 5-10 cm of the catheter can have a durometer of about 35 D.

(55) Any surface of various components of the system described herein or portions thereof (e.g., **110**, **120**, **130**, **140**, **150**) can be provided with one or more suitable lubricious coatings to facilitate procedures by reduction of frictional forces. Such coatings can include, for example, hydrophobic materials such as Poly TetraFluoroEthylene ("PTFE") or silicone oil, or hydrophilic coatings such as Polyvinyl Pyrrolidone ("PVP"). Other coatings are also possible, including, echogenic materials, radiopaque materials and hydrogels, for example.

(56) With continuing reference to FIG. 5, tether **140** has a proximal end **142** operably coupled to a hub **146**, or proximal anchor, that can be used to apply tension to tether **140**. Tether **140** further includes a distal end **144** that extends distally from passage **138** from the distal end of innermost catheter **130**. As depicted, hub **146** is removably coupled to carriage **240**, wherein sliding movement of carriage **240**) with respect to track or rail **201** results in proximal-distal movement of tether **140**. Preferably, hub **146** and carriage **240** are used in order to maintain tension in tether or rail **140** to facilitate use of the system **100**. For example, in some implementations, hub **146** can include a tension spring or elastic member that can be stretched to maintain tension in tether **140**. Likewise, if desired, any one of the carriages **210**, **220**, **230**, **240** may include an upper component that is slidably coupled to the main body portion (e.g., **210**) of the respective carriage. If so equipped, the carriage can similarly include a tension spring to cause the catheter carried by the

respective carriage to return to a desired axial location along the direction of rail **201**. This can be useful, for example, in the instance of catheter **130**, which can be spring loaded to return from a first deformed position to a contracted position between a first axial location and a second axial location to facilitate a reciprocating cutting operation.

(57) FIG. **6** illustrates further aspects of the catheter-based system **100** of FIG. **5** in accordance with some aspects of the present disclosure. For purposes of illustration, and not limitation, FIG. **6** illustrates a distal region of system **100** in accordance with some implementations of the disclosure. Illustrated are the distal end **114** of the outermost catheter **110**, the distal end **124** of the intermediate catheter **120** extending distally from the lumen **118** of the outer catheter, and the distal end **134** of the innermost catheter **130** extending distally from the lumen **128** of the intermediate catheter **120**. A distal region of tether **140** extends distally from the lumen **138** of the innermost catheter **130**. The distal region of tether **140** can include a plurality of markers **141** that are visible under fluoroscopy or other imaging modality (e.g., MRI and the like) that are set a predetermined, and known, distance apart from each other. A distal end **144** of tether **140** is operably coupled to a tissue anchor **150**. Tissue anchor **150** is anchored into tissue within a patient (discussed in further detail below) to permit tension to be applied to tether **140**. Once a suitable amount of tension is present in tether **140**, tether **140** acts as a guide rail to guide the movement, for example, of the distal end region of innermost catheter **130** to facilitate a tissue cutting operation. The anchor **150** can be provided with a proximally located pivoting coupling **152** to permit the tether **140** to pivot with respect to the anchor **150**. The anchor **150** may be made from a radiopaque material, or may be provided with one or more radiopaque markers (not shown) or other markers, as desired. The markers (e.g., **141**) can facilitate in situ measurement of a LVOTO, as well as helping to confirm the distance over which catheter **130** reciprocates to perform a cutting procedure, including helping to defined the proximal and distal movement limits of catheter **130**) when a cutting element, such as an electrode, of catheter **130** is electrified. This can be used to control the length of the cut through the tissue. In some implementations, if desired, the tether **140** and anchor **150** may not be present and instead a microcatheter or needle (not shown) may be slidably received within lumen **138** of catheter **130**. Moreover, innermost catheter **130** can be provided with a retractable and/or fixed electrode, as well as one or more sensors for detecting the presence of myocardium, and/or to detect electrical signals in the myocardium. In accordance with further implementations, the markers **141** and/or the anchor can be electrodes that provide a return path for electrical current that is supplied by supply electrode(s) defined in the distal region or near or at the distal end of catheter **130**. If so equipped, tether **140**) can include a conductive core, such as if tether **140** comprises an electrically conductive insulated tether. Anchor **150** and/or markers **141** can be in electrical communication with the conductive core of the tether **140** to provide a return path for the electrical current. In any implementation herein, a separate catheter, such as a pigtail catheter, can be provided an introduced alongside the system (e.g., **100**) to provide a return electrode to provide a return path for electrical current and/or to control the direction of the flow of electrical current in a region of interest where a procedure is being performed on tissue of a patient. The cutting electrode and the return electrode are preferably moved together in tandem. The return electrode preferably has an enlarged surface area to reduce the current density at the surface electrode and to reduce ohmic heating of surrounding tissue to reduce or prevent undesired ablation and thermal effects.

(58) In further accordance with the disclosure, the electrodes (e.g. supply, return, sensing, and the like) used in the various embodiments disclosed herein can have any desired length or shape, and if desired, can have exposed lengths that can be varied, such as by extending an exposed portion of an electrode outwardly through a port defined in an electrically insulating tubular member. The electrodes can have shapes or surface features configured to concentrate electrical charge and current density as desired. The catheters can be configured to direct flush fluid over or adjacent to electrodes to help aid in cooling the electrodes, and to help avoid the clotting of blood, and to enhance cutting, where appropriate. Any embodiment disclosed herein can be operated in a

monopolar mode, or a bipolar mode, as desired, wherein the return electrode can be built into the catheter or an adjacent (e.g., pigtail) catheter to provide a return path for electrical flow.

(59) FIGS. 7A-7C illustrate still further aspects of the catheter-based system of FIG. 5 in accordance with some aspects of the present disclosure. For purposes of illustration, and not limitation, FIG. 7A illustrates a close-up view of a distal region of catheter **130** that protrudes from the distal end of intermediate catheter **120**. FIG. 7B depicts a close-up view of the tissue anchor **150**, and FIG. 7C illustrates placement of one or more electrodes **137** in the distal end region of catheter **130** for various purposes. Electrodes **137** are electrically coupled to one or more elongate conductors (not shown) that extend through the body **133** of catheter **130** to permit the conductors to be externalized from the patient to be coupled to diagnostic equipment and/or an electrosurgical supply.

(60) FIG. 7A depicts a distal end region of the system **100**, illustrating a distal end of catheter **120** from which a distal region of catheter **130** extends. As depicted, the catheter **130** can be a steerable catheter with active steering. The distal end **134** of catheter **130** can include a tapered region. One or more electrode(s) **137** can be present near the distal end of the catheter **130**. The electrodes herein (e.g., **137**, **137'**) can function as one or more of supply electrodes, return electrodes, and sensing electrodes. Rail **140** can define a plurality of markers **141** along the length of rail **140** in the region of where tissue is to be measured and cut. Markers **141** can be used under visualization in situ to measure the distance of tissue to be cut. The starting point for the cut and the ending point for the cut can be noted by the user. In some implementations, the markers **141** can be configured as electrodes that can function as one or more of supply electrodes, return electrodes, and sensing electrodes. If desired, stops (not shown) can be placed proximate actuator/handle **136** to limit the proximal-distal range of the distal end region of catheter **130**. This can be done to define the length travel of the electrode **137** as it cuts through tissue. A distal marker **134** can be provided at the distal end **134** of catheter **130** in order to optimize positioning of the distal end **134** of catheter **130**.

(61) With reference to FIG. 7B and FIG. **14**, the anchor **150** may include a pivoted coupling **152** that is pivotally coupled to a distal end **144** of the tether **140**. As depicted, tether or rail **140** terminates at its distal end in a loop **149** that can be mated with loop **152** to provide a swiveling coupling to permit a large degree of rotational travel of the tether **140**, particularly once the anchor **150** is anchored into tissue. The anchor can include one or more tines that can be anchored into tissue. Anchor **150** as depicted includes three tines **154**. Anchor **150** is preferably formed from a shape memory material, such as a nickel-titanium alloy having shape memory formed therein such that the tines **154** of anchor **150** curl around to point proximally such that when tension is applied to the tether **140** inside a patient's heart, one or more of the tines **154** will be pulled into myocardial tissue, anchoring the anchor in place. In some implementations, anchor **150** can be made of a NiTi alloy, as indicated, and/or drawn filled tubing (DFT) that includes, for example, a NiTi alloy combined with a radiopaque material such as Platinum for visibility. Delivering the anchor **150** can be accomplished by pushing the anchor **150** to a target location confined within a distal end region of catheter **120**. Catheter **130** can surround tether **140**) and the distal end **134** of catheter **130** can push the anchor **150** out of the distal end of catheter **120** to deploy the anchor **150**. Alternatively, anchor **150** can be withdrawn into the lumen **138** at the distal end of catheter **130** and be deployed, for example, by advancing a small profile push rod down lumen **138** to push anchor **150** out of catheter **130**. Once unconstrained by the lumen **128** of catheter **120**, the shape memory configuration of the anchor **150** can cause the anchor **150** to deploy. The swiveling or articulating coupling attaching the tether to the anchor can control direction of the tines **154**. One or more markers **153** can be provided on the anchor near the pivot loop **152**, and/or on the tines **154** to help inform a user when the anchor tines anchor in tissue under a visualization modality. The catheter **130** can be articulated to push into the myocardium such that pushing the anchor out of the catheter causes the anchor **150** to penetrate the myocardium and become anchored in the tissue. In accordance with further implementations, any anchor herein (e.g., **150**, **350**) can be formed in a

flattened configuration. For example, rather than the three-dimensional hook implementation of FIG. 14, the tines can be formed from a planar material located adjacent widthwise with respect to each other along the planar material, and with one or more single direction hooks (with varying length and curvature) to make sure all hooks are embedded into tissue.

(62) In some aspects, the disclosure provides sensing catheters, and/or sensing catheters that can perform additional operations such as cutting. For example, the disclosure provides implementations of a medical device having an elongate body having a proximal end and a distal end, and an elongate tether operably coupled to the elongate body. The elongate tether and elongate body are configured to be longitudinally displaceable with respect to one another. The device further includes an electrode disposed on at least one of the elongate body and elongate tether, and electrical circuitry operably coupled to the electrode, wherein the electrical circuitry is configured to determine a state of at least one of the medical device and the anatomical tissue.

(63) In accordance with further aspects, the electrical circuitry, which can include one or more analog and/or digital circuit components, such as programmable processor programmed with machine readable code, which may be contained in a memory or other non-transient medium, can be configured or programmed to carry out the functions of the circuitry, wherein the term circuitry is intended to encompass analog and/or digital circuitry. In some implementations, the circuitry can detect an incoming signal from the anatomical tissue to confirm that the electrode (e.g., 137, 141) is in physical contact with the anatomical tissue. The incoming signal from the anatomical tissue can include an electrocardiogram signal from cardiac tissue. In further implementations, the electrical circuitry can be configured to detect a voltage or current drop across the electrode after electrical power has been applied to the electrode.

(64) In some implementations, the electrode(s) can be configured to cut tissue when energized, and relative longitudinal movement of the elongate tether and the elongate body can cause the electrode to cut through anatomical tissue that the device is placed adjacent to when the electrode is energized. In such applications, if desired, the electrical circuitry can be configured to correlate the voltage or current drop with a physical state, such as (i) a state of tissue being cut by the electrode, and a state of fouling of the electrode. If desired, such a system may further include a one or more pressure sensors located proximate the electrode. The pressure sensor can be operably coupled to the circuitry, for example. The processor can be programmed, for example, to determine at least one biological parameter based on receiving a signal from the pressure sensor, such as blood pressure or other parameters. The pressure sensor can include any suitable pressure sensor, such as a fiber optic pressure sensor having a distal crystal configured to direct light out of the distal crystal. The characteristics (e.g., wavelength or frequency) of the reflected light can permit computation or estimation of a local fluid pressure. Such a fiber optic sensor may be slidably received within the system (e.g., 100) and may be deployed to a target location to measure local pressure. Such a blood pressure probe may have one or more visualization markers to facilitate visualization of the blood pressure sensor under an imaging modality such as fluoroscopy or MRI. In another implementation, a pressure tap (opening) can be located at a desired location (e.g., a side port or distal opening of a catheter) that is in communication with a passageway through the device (e.g., 100) to an external pressure sensor, wherein static pressure at a predetermined location (e.g., a side port or distal opening of a catheter) can be measured. A pressure sensor can be deployed to help a user understand if a catheter is located distally beyond the aortic valve and is within the left ventricle of a patient. Such a sensor can also help a user appreciate whether or not an electrode is proximal a node, such as the SA or AV node, to avoid inadvertently delivering electrical current to a node.

(65) With reference to FIG. 7C, the electrode(s) can be retractable into a further lumen defined in the tubular member 133 (not shown) or can be fixed in place. Such electrodes 137 can be distributed about part or all of the circumference of the distal end region of catheter 130. Moreover, instead of a cutting electrode, a blade can be provided that can be fixed on the surface of catheter

130 that can be retracted into the distal end of catheter **120**. If a blade is provided, it can further be retractable into catheter **130**. The cutter can thus include, for example, a cutting electrode, a blade, a cutting wire such as a diamond wire, a wire with a cutting implement coupled to the cutting wire, a narrow gage metallic wire, and the like. The electrode(s) can sit proud of the surface of catheter **130** (e.g., protrude slightly when unconstrained) but can flex down to surface height to reduce profile when being delivered through the lumen **128** of catheter **120**. One or more electrodes **137A-137N** can be provided along the length of and/or about the periphery of tubular member **133**. The electrodes herein (e.g., **137A-137N**) can function as one or more of supply electrodes, return electrodes, and sensing electrodes. If used as cutting electrodes, the electrodes **137A-137N** can be fired at different times or simultaneously, as desired. For example, a first row of electrodes can function as supply electrodes, and a second row of electrodes can function as return electrodes. Tissue disposed between the supply and return electrodes of FIG. 7C, as well as in any other bipolar arrangement disclosed herein, can function as a conductive medium to complete the circuit. Some electrodes can be dedicated for cutting, and some of the electrodes can be dedicated for other purposes, such as sensing electrical properties of tissue, such as myocardial tissue. For example, some electrodes can be provided that is an EKG sensing electrode. Moreover, one or more of the electrodes can have more than one state of operation. For example, one or more of the electrodes **137** can be configured to sense electrical signals or properties of the tissue they are adjacent when power is not applied to the electrode to cut tissue. Additionally, one or more of electrodes **137** can be supply electrodes that supply current, and one or more of electrodes **137** can be electrodes that return current to an electrosurgical generator. In a further implementation, the anchor **150**) and/or markers can function as return electrodes as mentioned above. The system, if utilizing one or more electrosurgical cutting electrodes can operate in a monopolar state of operation, wherein the electrode(s) are supply electrodes, and the return path of the current to the electrosurgical generator is through the patient by way of a conductive pad attached to the patient. Alternatively, the system can operate in a bipolar mode of operation wherein one or more electrodes can be supply electrodes and one or more electrodes can be return electrodes such that the electricity only need to pass from a supply electrode to a return electrode to complete a circuit. Suitable electrical supply and return conductors can be provided extending through additional lumen(s) of catheter **130** or along a core conductor (not shown) of tether **140** for example that are externalized and operably coupled to an electrosurgical generator and/or diagnostic equipment, as desired.

(66) FIGS. **8A-8F** illustrate features of the catheter-based system of FIG. **5** to confirm device axial and rotational orientation while performing a procedure in accordance with the present disclosure. FIG. **8A** depicts a specially configured marker **139** that, when integrated into a catheter, permits a user to determine the rotational orientation of catheter **130**. Moreover, an instrumentality, such as an electrode **137** and/or an exit port for a movable electrode can be positioned diametrically opposite of the marker **139**. FIG. **8B** depicts an implementation of a catheter **130** wherein a cutter, comprising a cutting electrode, or a blade, or an ultrasonic transducer or an ultrasonic element coupled to a cutting element **137** is provided in a distal end region of catheter **130**. Where used, a cutting electrode may be coupled to an RF power supply operating in normal regimes, or microwave regimes that are less likely to interfere with the beating of the heart or the operation of cardiac nodes. An ultrasonic cutter provides advantages such as reduced tissue heating and not interfering with electrical operation of the heart, or ablation procedures. An ultrasonic transducer can be operably coupled to a cutting blade to effectuate cutting in accordance with any implementation of the present disclosure.

(67) FIG. **8C** depicts the implementation of FIG. **8B**, but with marker **139** added. The marker **139** can also function as a return electrode for bipolar operation when attached to a current return line (not illustrated). The shape of the marker **139** includes an elongate spine section coupled to a mid-point of a “C” shape at each end of the spine, such that the “C” shaped portions wrap around a portion of the circumference of the catheter **130**, above or beneath the surface of body **133**. If the

orientation of the marker **139** is known to the user to be diametrically opposite the electrode **137**, the user can rotate the catheter **130** until the electrode is in contact with a surface to be cut. If the electrode **137** is configured to sense electrical activity in the tissue, the user can be informed that the electrode **137** is in electrical contact with myocardium, for example, if the electrode **137** conducts electrical signals from the myocardium to a signal detector or processor configured to detect such electrical signals. FIG. **8D** depicts a deployable electrode **137** that is withdrawn into a lumen defined in body **133** of catheter **130**. When the electrode is deployed as depicted in FIG. **8E**, the location where the electrode exits is indicated by the location of marker **139**. FIG. **8F** is a view of the marker **139** under fluoroscopy demonstrating the electrode **137** being advanced out through a side opening of the catheter **130**.

(68) FIGS. **8G-8P** depict aspects of a further implementation of a distal region of an inner catheter **2130** in accordance with the present disclosure that can be used in place of catheter **130** described above. FIG. **8G** is a side view of the distal end region of the catheter **2130**, FIG. **8I** presents a cross-section of the distal end region of catheter **2130** down a vertical longitudinal centerline thereof and FIG. **8H** presents a wireframe view of the distal end region of catheter **2130**. FIG. **8J** presents an isometric view of the distal end region of catheter **2130**, whereas FIG. **8K** presents the same view with a distal tubular covering **2130C** removed to illustrate the relative location of interior components.

(69) Catheter **2130** can be coupled at a proximal end thereof to a suitable actuator, including but not limited to those described elsewhere in this disclosure with reference to FIGS. **31-40D**. The inner catheter **2130** is built around an inner tubular member **2130E** that defines a lumen **2130A** along at least a portion of its length (all or a part of its length). An electrode **2131** is concentrically positioned about and coupled to a distal end region of inner tubular member **2130E**. The electrode **2131** is electrically coupled at a proximal end thereof to an electrical conductor, such as an electrically insulated or uninsulated elongate metallic member that extends proximally to a proximal end of catheter **2130** and operably coupled to an electrosurgical power supply (not shown). Inner tubular member **2130E** may be partially or fully polymeric, or can include a braided layer of polymeric or metallic braids in a coextrusion with another material, and/or may include a polymeric or metallic coil, a lasercut hypotube to enhance flexibility, and the like.

(70) The lumen **2130 A** can have an anchor **2150** and extension rod **2140** slidably disposed therein as with other implementations herein. As illustrated, proximal movement of the anchor **150** can be limited by a stop **2139** in the form of a boss, bump, protrusion, or as illustrated, a tubular member attached to an inner surface of tubular member **2130E** through which rod **2140** is slidably disposed, or may be attached as a tubular member outside tubular member **2130E** that applies an inward radial force to compress or restrict the inner diameter of tubular member **2130E**. The catheter **2130** further includes a flush lumen **2135** that can be defined by a tubular member, for example that is disposed alongside and parallel to, or if desired concentric with, the tubular member **2130E**. As depicted, the flush lumen **2135** is disposed diametrically displaced from, or opposite to, the conductor **2133** about the tubular member **2130E**.

(71) As further depicted, a pull wire **2161** is attached at a distal end thereof to a distal tip region of catheter **2130**. As illustrated in FIG. **8L**, the pull wire **2161** is attached, such as via welding, adhesive, or crimping to a distal retainer, anchor band or crimp **2165** having a “C” shaped or circular cross section. Distal retainer **2165** is attached directly or indirectly to the distal region of tubular member **2130E**. For example, a sleeve **2169** with a longitudinal slit may be provided wherein the slit accommodates lumen **2135** and steering wire **2161**, but contacts an inner cylindrical surface of anchor **2165** and an outer surface of tubular member **2130E**. Outer shaft **2130C** can be made from braided or other suitable material to facilitate structural deflection.

(72) The pull or steering wire **2161** extends proximally into a distal end of a tubular member **2167** that is in turn operably coupled to a proximal region of the catheter (in a location of the distal end region of the catheter proximal to the distal retainer **2165**). As illustrated, a distal location of

tubular member **2167** is attached to a retainer or anchor **2163** that can be (but need not be) in turn directly or indirectly coupled to tubular member **2130E**. In this manner, when tension is applied to the pull wire **2161**, the catheter tip deflects in the direction of the tension to help steer the distal end of catheter **2130** through tortuous anatomy. A distal tubular covering **2130C**, which may be a continuation of a braided shaft **2130D** having a radiopaque polymeric (e.g., PEBAX) outer jacket and liner surrounds components **2130E**, **2133**, **2135** and **2161** that transitions to a proximal tubular covering **2130D** that preferably has a suitable durometer to complement surrounding anatomy and facilitate deflection of the distal region of the catheter under tension of the steering wire **2161**.

(73) With reference to FIGS. **8P** and **8L**, the electrode **2131** is formed from a tubular section **2131B** that surrounds the distal tip region of the tubular member **2130E**. One or more windows **2131D** can be defined through tubular section **2131** to provide additional contact perimeter for adhesive to facilitate coupling of the electrode **2131** to tubular member **2130E**. Windows **2131D** in combination with the dorsal fin **2131A** and connecting structure, if formed from radiopaque material, or material that can be displayed on another imaging modality, such as MRI, can function as a three dimensional marker to help a user determine longitudinal, rotational and angular displacement and movement. In particular, windows **2131D** can act as alignment holes such that when they are aligned with each other under visualization a user can conclude the orientation of the tip of the device with relative certainty. In addition, the overall shape of the structure can have a “P” shape further aiding in alignment.

(74) Electrode **2131** further includes a proximally extending section **2131C** that provides contact with conductor **2133** to facilitate electrical and structural attachment between the two components. The section **2131C** can be a partial circumferential continuation of tubular section **2131B**. Electrode **2131** further includes a dorsal fin **2131A** that extends radially upwardly from tubular section **2131B** and is oriented in a manner that is axially aligned with the exit port **2135D** of the flush lumen. Electrode **2131** is preferably coated in a dielectric coating **2131F** that surrounds tubular section **2131B** and both planar sides of the dorsal fin **2131A**, but that leaves some or all of the edge of dorsal fin **2131A** electrically exposed. In this manner, when the electrode **2131** is energized, the electrical charge is concentrated at this edge surface to provide more efficient distribution of electrical power, and also helps ensure that the electrically energized surface is within the flow stream of fluid exiting port **2135D**. For example, a dextrose solution or other solution can be flushed through the flush lumen **2135** and over the fin **2131A** of electrode **2131**. This can keep blood cells away from the electrified surface when the electrode is energized and, in the case of dextrose or other non-conductive fluid concentrate electrical current to flow to tissue rather than fluid surrounding tissue, thereby reducing the chances of formation of thrombus and associated blood clots.

(75) FIGS. **8M-8O** more fully illustrate the anchor **2150** and pusher rod or tube **2140** that is slidably disposed in a distal end region of lumen **2130A**. Anchor **2150** is similar in many regards to anchor **150** described elsewhere herein. FIGS. **8N** and **8P** illustrate the anchor tines **2154** in a constrained condition as when they are pulled proximally into lumen **2130A**, and FIG. **8M** illustrates the tines **2154** after deployment. The relative location of stop **2139** is indicated in FIG. **8O**, with the understanding that the stop **2139** is coupled to the inner surface of tubular member **2130E** within lumen **2130A**.

(76) Electrode **2131** can be a monopolar electrode and the system can include a return electrode as a separate catheter (or a standard return pad electrode located elsewhere) disposed near the cutting location to provide a return path for electric current. Alternatively, the catheter **2130** can be provided with one or more return electrodes (not shown) located near the electrode fin **2131A**. For example, such a return electrode can be located on a dorsal surface of the catheter **2130**, such as proximal to the exit port **2135D** of the flush lumen **2135**, or other suitable location.

(77) FIGS. **9A-12C** illustrate aspects of a representative procedure to treat a LVOTO in accordance with some aspects of the present disclosure. For purposes of illustration, and not limitation, FIGS.

9A-9C depict the distal region of system **100** from different angles. As depicted, outermost catheter **120** deflects over the aortic arch to the ascending aorta. Catheter **120** facilitates deflection just under the valve alongside a base side of the left ventricular outflow tract obstruction (“LVOTO”). As depicted in **FIGS. 10A-10B** catheter **130** holds the distal end portion of catheter **130** bearing an electrode and anchor **150** and is deflected over the LVOTO in a controlled manner to deliver the anchor **150** to a location at the base of the distal end of the LVOTO toward the apex of the left ventricle. As depicted in **FIG. 10C**, tension is then applied to tether **140** and catheter **130** to conform catheter **130** to the shape of the obstruction.

(78) **FIGS. 11A-11C** depict aspects of a cutting operation in accordance with the present disclosure utilizing a reciprocating cutter. For purposes of illustration, and not limitation, the innermost catheter **130** is retracted proximally and advanced distally in a reciprocating manner while the electrode **137** is energized and in contact with the myocardium forming the obstruction. **FIG. 11A** depicts the innermost catheter when the distal end thereof bearing the cutting electrode (not shown) is about halfway along its reciprocating movement while traveling along tether rail **140**. As depicted in **FIG. 11B**, markers **141** along the length of tether or rail **140** can be provided to permit a user to measure the cut length, and thus the linear travel that the actuator/handle **136** of the innermost catheter has to traverse along the proximal region of the tether **140**. The linear range of motion of the actuator **136** or the carriage **230** can then be fixed, for example, by placing a clamp on either side of carriage **230** to prevent carriage **230** from moving proximally or distally by too great of an amount. The distal end of catheter **130**, now functioning as a cutter due to electrode **137** being energized, reciprocated back and forth. With each movement, the electrode **137** burns further through the tissue of the obstruction until the cutter bottoms out along a line defined by the pathway of the tether between the anchor **150** and the distal end **124** of the intermediate catheter **120**. The maximum depth of the cut is therefore controlled by placement of the distal end **124** of the intermediate catheter **120** and the pivot **152** of the anchor **150**. **FIG. 12A** depicts advancing the innermost catheter **130** along a distal direction over tether/rail **140** to cause it to bow outwardly into the left ventricle until it contacts the anchor **150**. The tether **140** is then withdrawn proximally into the distal end **134** of catheter **130** as depicted in **FIG. 12B**. The catheters **110**, **120**, and **130** may then be withdrawn proximally as indicated in **FIG. 12C** leaving behind the LVOTO which has been flayed with a cut, opening up the LVOT. If necessary, prior to removing the system **100**, a second cut can be made through the LVOT that is substantially parallel to the first cut by realigning the anchor and redeploying the anchor, or by shifting the distal end **124** of the intermediate catheter **120**. **FIGS. 13A-13C** illustrate results of an illustrative LVOTO treatment procedure in a porcine model in accordance with some aspects of the present disclosure. **FIG. 13A** depicts an elongate cut made in the obstruction by system **100**. Also illustrated are the relative locations of the LV apex and the aortic valve, which is undamaged by the procedure. As depicted in **FIG. 13B**, the cut is over 5 mm deep. And, as depicted in **FIG. 13C**, the cut is nearly 40 mm long.

(79) In further accordance with the disclosure, **FIGS. 15-21B** illustrate a further implementation of a tissue cutting system in accordance with the present disclosure.

(80) For purposes of illustration, and not limitation, **FIG. 15** illustrates a multi-catheter system **100'** wherein like reference numerals are directed to analogous system components as system **100** described above. System **100'** is illustrated as having the same basic catheter arrangement as system **100**, with an outermost catheter **110'**, an intermediate catheter **120'** and an innermost catheter **130** that can be actively steerable as with implementation **100**. While system **100** set forth above describes use of an inner tether or rail **140**, system **100'** instead includes an inner push tube **140'** that extends along a passageway defined through catheter **130'** that includes an anchor **150'** disposed at a distal end thereof. System **100'** can be made of similar materials and dimensions as system **100**. The anchor **150'** pivots at the distal end **144'** of the push tube **140'** to allow the push tube **140'** to lay against and along myocardial tissue, such as the septum, as a guide rail in a manner similar to rail/tether **140** discussed with respect to implementation **100**.

(81) Innermost catheter **130'** can include a ring-shaped electrode **137'** that is configured to cut tissue to which it is adjacent when it is energized. One or more supplemental electrodes **137A'** can be provided that can sense electrical parameters of myocardial tissue and/or act as a return path for electrical current to facilitate bipolar operation of system **100'**. Similarly, push tube **140'** or a conductor disposed therein (not shown) can serve as an electrically conductive return path to facilitate bipolar operation. The electrode **137'** at the distal tip of catheter **130'** may be of a fixed depth to cut a predetermined amount, or may be adjustable to permit the cutting depth to be adjusted. The electrode **137** may further be operably coupled to a hollow needle disposed within catheter **130** to permit access of a movable electrode or other wire.

(82) System **100'** can further be configured to permit a blood pressure reading to be taken from within lumen **128'** of intermediate catheter **120'** to confirm placement of the distal end of catheter **120** below the aortic valve of the patient. Moreover, with regard to any catheter system set forth in the present disclosure, a fluid channel can be provided to deliver a beneficial agent, such as a flush (e.g., a dextrose flush) to the area being treated. Such fluid can be delivered through the annular space defined between an inner wall of catheter **130'** and an outer surface of push tube **140'**, as desired, or by another route or separate catheter.

(83) With continuing reference to FIG. **15**, the intermediate catheter **120'** can further include a deployable anchor wire **127'** to facilitate anchoring the distal end of catheter **120'** in a desired location, such as at the base of the LVOTO. In some implementations, the anchor wire **127'** can be deployed distally from a lumen defined in the distal end of catheter **120**. In other implementations, the anchor wire can be deployed from a location of the innermost catheter **130**. The anchor wire **127'** can comprise a tubular member, such as a hypotube that can permit a guidewire or other wire, such as an electrode. In other implementations the anchor wire **127'** can comprise a wire pigtail-like loop to sit in an aortic valve cusp.

(84) FIG. **16** is a schematic of anatomy of a heart illustrating a target cutting site for treating a LVOTO in accordance with the present disclosure. Specifically, the schematic illustrates a cross-section of the heart illustrating the relative locations of the left ventricle, right ventricle, left atrium, aortic valve, and LVOTO, indicating the boundaries of the LVOTO with respect to what would be considered a normal septal wall. FIGS. **17A-17B** are schematics illustrating aspects of introducing a system into a patient's heart to treat a LVOTO in accordance with the present disclosure. Specifically, a guidewire **160'** is introduced into the left ventricle (e.g., a 0.035-inch wire), and a dilator catheter **170'** is introduced over guidewire **160'**. Catheters **110'** and **120'** can then be introduced over the dilation catheter **170'** to deliver the distal ends of catheter **110'** proximal to the aortic valve, and catheter **120'** proximate the tissue mass to be cut with the system **100'**.

(85) FIGS. **18A-18B** are schematics illustrating further aspects of deployment of system **100'**. FIG. **18A** depicts anchor **150'** deployed into the myocardium with the push tube **140'** surrounded by the distal regions of catheters **130'** and **120'**. If needed, catheter **120'** can be advanced to the site of the anchor **150'** to support the deployment of the anchor **150'**. FIG. **18B** illustrates the system after the intermediate catheter **120** has been retracted proximally over the obstruction a location near the base of the obstruction at the proximal side of the obstruction, and indicating the deployment of the anchor **127'** into the tissue of the mass. Also illustrated is an alternative placement of anchor **127** in the cusp of the aortic valve wherein the anchor **127'** acts as a brake that pushes against the cusp of and that gets trapped to an extent in the aortic valve cusp.

(86) FIGS. **19A-19C** are schematics illustrating further aspects of use of system **100'** to treat a LVOTO in accordance with the present disclosure. FIG. **19A** depicts deployment of a dynamic electrode **137'** that may be advanced distally and radially outwardly with respect to the distal end of catheter **130**. The catheter may alternatively be a fixed or static electrode **137'** as indicated in FIG. **19B**. As a dynamic electrode **137'**, the depth of the cut can be adjusted by advancing more or less of the electrode **137'** from the catheter **130'**. To complete a pass of cutting tissue, the catheter **130'** is then withdrawn proximally over the push tube **140'** which functions as a guide rail. One or a

plurality of passes may be performed to cut the tissue multiple times along the same line of cutting to obtain a sufficiently deep cut to open the LVOT to an adequate degree.

(87) FIGS. 20A-20B are schematics illustrating further aspects of system **100'** and its use to treat a LVOTO in accordance with the present disclosure including resheathing the cutter and retrieving the anchor of the system. FIG. 20A depicts the innermost catheter **130')'** being advanced distally over push tube **140'** to the location of anchor **150'**. As illustrated in FIG. 20B, the push tube **140'** is withdrawn into the distal end of catheter **130'**. FIGS. 21A-21B depict withdrawing the catheter-based system from the patient and illustrating the treated LVOTO. FIG. 21A depicts the distal portion of the catheter system **100'** as it is withdrawn from the patient's heart. FIG. 22 depicts the bottom of the cut through the LVOTO, which helps open a channel to enlarge the cross-sectional area of the LVOT. If it is desired to make the cut deeper, or to make a second cut in the tissue generally parallel to the first cut, the system **100'** can be re-deployed into the left ventricle, the anchor **150'** can be re-anchored in a laterally displaced location from the first site of deployment of anchor **150'**, and a second cut can be performed in a manner similar to the first cut.

(88) FIGS. 22A-23C illustrate aspects of a further technique to treat a LVOTO in accordance with the present disclosure utilizing a further variation of a catheter-based system **100''**. System **100''** includes an inner rod or tubular member **140''** that forms a main body of the system about which other system components are arranged concentrically. An intermediate anchoring catheter **120''** is slidably disposed about member **140''** to permit catheter **120''** to slide proximally and distally over member **140''**. A second, distal catheter **180''** is disposed distally of catheter **120''** and also disposed slidably about member **140''**. In embodiment **100''**, the catheter system **100''** is anchored in place at the distal base of the obstruction (e.g., an LVOTO) to hold the system in place to perform a cut along the obstruction at a location that is proximal to the anchor. As illustrated, the system **100''** is deployed into the left ventricle, wherein the distal catheter **180** is advanced distally past the obstruction toward the apex of the left ventricle. The distal end region of the catheter **120''** is advanced to the base of the obstruction, and an anchor wire exit port is "aimed" at the distal base of the obstruction. An anchor wire **127''** is similar to anchor wire **127'** can be advanced proximally into the distal base portion of the obstruction and pulled proximally into the tissue mass. In the implementation of FIGS. 22A-22C, this is effectuated by pulling outer catheter **180''** proximally, which is attached to the anchor wire **127''**. This pushes the anchor wire **127''** proximally into the tissue mass. At this point in the procedure, catheters **120''** and **180''** are maintained in position by the anchor wire **127''**. This permits tension to be applied to the system to push the catheter system against the tissue mass.

(89) With reference to FIGS. 23A-23C, an outer catheter **110''** is slidably disposed about intermediate catheter **120''**. An elongate cutting electrode **137''** extends alongside outer catheter **110''**, wherein articulating, or steering the catheter **110''** or applying tension to the cutting wire while the catheter **110''** is bent over the tissue mass causes the electrified wire **137''** to urge against the tissue. When energized, electrode **137''** completes a circuit through the tissue mass in a monopolar manner, or a bipolar manner, wherein a bipolar circuit can be completed through anchor wire **127''**. The outer catheter **110''** can be reciprocated along a proximal-distal direction over intermediate catheter **120''** to complete the cutting procedure. The system **100'** can then be removed by pushing catheter **180''** distally over core member **140''**, causing the anchor wire **127''** to be withdrawn into catheter **120''**. The system can then be removed, leaving behind the obstruction that has now been cut along its length to open the LVOT channel. While not specifically illustrated, the system components may be surrounded in a proximal location in the region of the aortic valve by an outer deflectable catheter to protect the aortic valve from electrode **137''**.

(90) FIGS. 24A-25C illustrate aspects of still a further technique to treat a LVOTO in accordance with the present disclosure that includes deploying a further implementation of an anchor. As illustrated, rather than using an anchor similar to anchor **150** or **127''** as set forth above, further implementations of an anchor (**350**, **450**) are described that can be deployed out of a distal end of

an innermost catheter (e.g., **130**, **130'**) to anchor a rail (e.g., **140**, **140'**) in place to facilitate a reciprocating cutting operation through a tissue mass (e.g., LVOTO) as discussed elsewhere herein. (91) FIGS. **24A-24C** illustrate an anchor **350** that is pivotally coupled to a distal end of an elongate inner member **340**, such as a tubular member about a pivot point **345**. It is preferable that the elongate inner member **340** includes sufficient pushability to push anchor **350** distally out of a hypotube (e.g., **140'** in FIG. **15**), polymeric or composite tubular member, or the like. Inner member **340** can be deployed distally from a distal end **334** of a sheath or innermost catheter, for example. A tension member **344**, such as a tether, can be coupled at a distal end to the anchor **350**, and may be directed proximally into an exit port **342** of inner member **340** into a lumen thereof, and be externalized from a patient. FIG. **24A** depicts the anchor **350** being deployed from the sheath or inner catheter, wherein a free distal, tissue piercing end of the anchor **350**, which may be a flat wire or other shape, contacts the tissue along a direction indicated by the arrow. Tension is applied to the tether **344** causing it to shorten and cause anchor **350** to articulate about the pivot point **345** until the tine(s) of the anchor **350** begin to point along a proximal direction as indicated by the arrow in FIGS. **24B** and **24C**. In some implementations, tether **344** can include a pushable wire that can both be pulled proximally and pushed distally to help pivot the anchor about the pivot point **345** to deploy, and retrieve the anchor **350**. When the anchor **350** is sufficiently proximally directed, the inner member **340** can be pulled proximally, anchoring the tine(s) of the anchor **350** into tissue, such as at the distal base of an obstruction such as a LVOTO. FIGS. **2A-25C** depict an alternate version of anchor **450** that is also configured to be pulled by a proximally tensioned tether (not shown) to cause the anchor **450** to articulate about pivot **445**. Tension can then be applied to the inner member of the system, pulling the tines of the anchor **450** along a proximal direction into the tissue mass.

(92) FIGS. **26-30B** illustrate aspects of a further illustrative system **500** and associated method to cut tissue in accordance with the present disclosure.

(93) FIG. **26** depicts a catheter-based system encased in a sheath **500**, or outer steerable catheter, that can surround the system as it is delivered to the left ventricle of a patient. The outer sheath **510** has a distal end that abuts a soft distal tip **524** of the system. The distal end of any system herein can be rounded or tapered, as desired. An inner lumen of system **500** can accommodate a guidewire **560**. The guidewire **560** is first advanced to the left ventricle, and the system **500** can then be advanced over the guidewire. System **500** is sufficiently flexible to conform to the inner surface of the left ventricle as depicted in FIG. **26**. The outer catheter **510** may be deflectable and/or preset with a curvature to facilitate delivery and later control of the cutting system operation, discussed in further detail below. The outer catheter **510** can be retracted proximally to a desired extent to expose a cutting wire **537**, discussed in further detail below. As depicted in FIG. **26**, the system **500** is situated within the left ventricle such that a primary curve is formed in the system in the apex of the left ventricle and the distal end portion of the system is situated between papillary muscles on the far wall of the ventricle. The system further conforms to and bends around an obstruction in the LVOT.

(94) FIG. **27** depicts a further step in the procedure of using system **500** to form a cut in the LVOT. In this state, the outer sheath **510** has been retracted to the extent that the distal end **514** of the outer sheath is proximal to the tissue obstruction, but distal to the aortic valve to provide protection to the aortic valve during the cutting procedure. An inner catheter **520** is coupled to the soft distal tip **524** of the system. In addition, the inner catheter **520** defines a cutter exit port **537A** and a cutter entrance port **537B** that receive therethrough a reciprocating cutting element **537** therethrough. The cutting element **537** can extend distally into the body of catheter **520** through distal port **537B** and be anchored at a fixed location to a rigid structure, or to an elastic structure such as a tension spring or piece of elastic. The cutting element **537** can also extend proximally into the body of catheter **520** through proximal port **537A** to an actuator (not shown) that is external to the patient. When tension is applied to the cutting element **537** in a proximal direction, the cutting element is pulled

proximally against the force, for example, of a tension spring (not shown) disposed within the body of the catheter **520** located distal to distal port **537B**. When proximal tension is released, the cutting element **537** is pulled distally. Thus, repeated applications and removal of proximal tension to the cutting element **537** create a sawing action that cuts through the tissue either if the cutting element **537** is electrified, and/or is sharpened. Moreover, the cutting element **537** can include one or more discrete electrodes thereon that, when energized, cut through tissue as they are pulled along with the cutting element. If desired, a distal anchoring wire **527** and associated exit port **527** can be provided, wherein the anchoring wire **527** can be advanced proximally into the distal base portion of the tissue mass to help stabilize the system **500** in place. If desired, one or more radiopaque markers (not shown) can be provided at various locations on system **500**, such as at the distal and **514** of the sheath, as well as at the exit port **537A**, entrance port **537B**, and the distal tip **524** of the system.

(95) A variety of aspects can be used to anchor and/or stabilize system **500** in place. For example, the inner catheter **520** can be placed in the apex of the left ventricle with the distal segment against the free wall of the ventricle between the papillary muscles. This stability can be enhanced with preset curves to fit the hypertrophic cardiomyopathy (HC) or obstruction on the septum. Distal and/or proximal to the cutting target, an exit port may allow an anchoring/stabilizing element to exit the shaft, such as element **527** mentioned above. This element may include, for example, a wire that crosses the myocardium into the right ventricle, a curved element that embeds in the myocardium, or a preformed element (e.g., from a NiTi alloy) that presents as stabilizer feet. A preformed curve may be formed into body **520** in the location where it crosses over the cutting target (LVOTO) to predispose the curve to pull the distal point of the curve closer to the base of the cutting target. This can be enhanced by shortening the cutting element from the proximal handle to apply tensile force to the region of the device **520** that is distal to the entrance port **537B**. The sheath **510** may be a deflectable catheter with the curve directed toward the proximal portion of the cutting target. This keeps the proximal portion of the arch close to the myocardium. The proximal sheath **510** may also help control the location of the proximal anchoring element relative to the cutting target and mechanism.

(96) With reference to FIG. **28**, the system **500** is depicted in the same side cross sectional view as FIGS. **26-27** wherein the exposed cutter **537** has penetrated tissue through various potential cutting mechanisms (e.g., reciprocating elements) that move back and forth along the tissue. A variety of reciprocating mechanisms may be used to effectuating cutting with cutting element **537**. For example, as mentioned above, distally located spring disposed, for example, in body **520** may be used to return the cutting element distally after the cutter is pulled proximally. Alternatively, both ends of the cutting element **537** may be connected to a proximal tension element, wherein the cutting element or tether attached thereto reverses direction about a transition point, such as the distal entrance port **537B** or a rounded boss located inside inner catheter **520** located distally to the distal entrance port **537B**. Thus, both ends of the tension element coupled to the cutter **537** can be externalized and each end can be pulled alternately to effect reciprocating cutting motion. In another implementation, a proximal end of the cutter **537** can be connected to a tube or rod (not shown) located within inner catheter **520** located proximally with respect to exit port **537A** that advances and retracts through inner catheter **520** the distal end of the cutter **537** is connected to a spring or other elastic element disposed within the inner catheter **520**. Cutting element **537** in the region where cutting is performed may include a smooth radiopaque wire or tether, a textured radiopaque wire or tether, a diamond coated wire, a radiofrequency (“RF”) wire or electrode operating in RF or microwave regimes, a length of razor wire, a cutting blade coupled to a reciprocating tether, an ultrasonic transducer or element coupled to an ultrasonic transducer, and the like.

(97) FIG. **29** depicts an isometric view of the implementations presented in FIGS. **26-28**. As depicted, once cutting is complete, the tissue of the LVOTO is splayed open as defined by the

perimeter **590** illustrated in FIG. **29**. This splaying open of the tissue enlarges the cross-sectional area of the LVOT, thereby reducing the risks associated with implanting an artificial valve, as mentioned above. The system **500** can then be removed, for example, by advancing the sheath **510** distally over inner catheter **520** and cutter **537** to protect surrounding tissue from the cutter **537** during withdrawal. The system **500** can then be withdrawn with the guidewire **560**.

(98) FIGS. **30A-30B** depict a variation of system **500** wherein the reciprocating cutter **537** includes a first electrode **537C** disposed thereon for RF cutting, and a second electrode **537S** mounted thereon for EDEN (Electrocardiogramadial Depth Navigation) sensing. EDEN sensing is based on detection of an EKG signal to detect tissue depth. Each electrode **537C**, **537S** is electrically coupled to a conductor that may pass inside of cutter **537** and extend proximally through system **500** and out of the patient. As the cutting is being performed by electrode **537C**, electrode **537S** picks up EKG signals that can be used to help estimate the current depth of the cut into the myocardial tissue.

(99) FIGS. **31-40** depict further implementations of a catheter system in accordance with the present disclosure, or aspects thereof, having a different actuator assembly than the system of FIG. **5**. For purposes of reference, the proximal and distal directions of the assembly are indicated in various figures.

(100) FIG. **31** is a side view of a catheter-based system **1100** to perform a percutaneous procedure to cut tissue in accordance with the present disclosure. While the system **1100** can utilize a reciprocating cutter as described elsewhere herein, the system **1100** can be configured with different end effectors to perform different procedures, and can be provided with any desired number of overlapping catheters. For example, while the illustrated implementation **1100** includes a pair of concentric catheters **1120**, **1130** surrounding a central tether **1140**, a third, fourth and/or fifth concentric catheter can be added surrounding catheter **1120**. By way of further example, the central tether **1140** may be omitted in the assembly and the system can be used as a two-catheter system. Alternatively, a smaller catheter, such as a microcatheter can be inserted inside the lumen of catheter **1130**, and a further actuator similar to or the same as actuators **1126**, **1136** can be used proximally of catheter **1130** and inside of catheter **1130**.

(101) As indicated above, system **1100** includes two concentrically disposed deflectable catheters including a radially outermost catheter **1120**, within which a second deflectable intermediate catheter **1130** is slidably received. A tether **1140** in the form of a flexible suture, a wire with column strength and pushability, or a pushable tubular member, or other instrumentality, such as a guidewire, a snare or the like is slidably received within catheter **1130**. Each deflectable catheter **1120**, **1130** can include one or more steering wires (FIG. **34**, **1121**) to which tension can be applied to selectively cause the distal end of each respective catheter **1120**, **1130** to bend in a preferential direction. References to steerability or deflectability of catheters herein can encompass active steering, such as with a tension wire, or steering using flexible or deflectable catheters of lower relative durometer. The distal region of one or more catheters may be configured to bend in a preferential direction. Rotating each respective catheter **1120**, **1130** about their central axis will cause the bent distal end of each respective catheter to point in any desired direction inside the patient. The catheter **1120**, **1130** can then be advanced distally into a desired anatomical structure.

(102) As depicted in FIG. **31**, each **1120**, **1130** is comprised of a tubular body having a proximal end coupled to an actuator **1126**, **1136** and a free distal end (not shown) in a manner that is similar to, or the same as, catheters **120**, **130**. Each catheter **1120**, **1130** can be moved axially and rotationally with respect to the other components of system **1100**. Each catheter **1120**, **1130** is held in relative position by a respective carriage **1220**, **1230** that is slidably received on a rail **1201** of a stand **1200** (see FIGS. **37**, **38**). Tether **1140** is likewise coupled to a tensionable anchor **1146** that is in turn operably coupled to a carriage **1240**. Each carriage **1220**, **1230**, **1240** can be axially translated between the proximal end **1202** and the distal end **1204** of the rail **1201**. In the illustrated implementation, one or more of the carriages **1220**, **1230**, **1240** can be locked in place with respect

to the rail **1201** by engaging a rail lock by rotating a lock handle **1225**, **1235**, **1245**. For example, the lock handle **1225**, **1235**, **1245** can actuate a clamping mechanism to clamp the actuator **1126** in place with respect to rail **1201**, for example, by advancing a threaded fastener against the rail **1201**, or by tighten a clamp that surrounds the rail **1201**, or by rotationally advancing a cammed fastener against the rail **1201**, among other options.

(103) With continuing reference to FIGS. **31-32**, actuator **1126** is formed from a first component including a main body portion **1222** that can partially or fully surround rail **1201** and slide proximally and distally over rail **1201** when the lock handle **1225** is disengaged. The actuator **1126** extends upwardly from the base portion **1222** into an upwardly extending portion or mount **1224**, illustrated in FIG. **31** as a hollow outer housing portion that extends upwardly into an upward housing portion that defines a cradle to receive the proximal end of catheter **1120** therein. A clamp or door **1224A** closes over the catheter, holding it in place, for example, by way of an interference fit or other suitable disengageable connection.

(104) With reference to FIG. **32**, as with the implementation of FIG. **5**, each catheter **1120**, **1130** is received along a vertical direction into an upwardly extending mount **1224**, **1234'** of a respective carriage **1220**, **1230**. More specifically, a handle portion **1226A**, **1236A** of each actuator **1126**, **1136** can be lifted out of each mount **1224**, **1234** when a removable cover **1224A**, **1234A** is removed, permitting the respective catheter and handle to be lifted out of the mount. In the implementation of FIG. **5**, the catheter is received in a fork shaped coupling. In the implementation of FIG. **32**, the proximal end of each catheter **1120**, **1130** is surrounded by a hub **1272** that is comprised of a gear **1252** and a flange **1272A-1272E** that defines a circumferential channel **1272E** about at least a portion of its periphery that straddles a ridge or boss **1229** (FIG. **33**) in the mount **1224**, and is rotatably received with respect to the ridge or boss **1229** so that the handle **1126A** may be rotated within mount **1224** when the knob **1254** and worm gear **1256** is rotated, causing gear **1252** to rotate. When installed about ridge **1229**, the catheter including handle portion **1226A** is held in place longitudinally in mount **1224** when door **1224A** is closed, but the handle and hub **1272** can rotate in place in the mount **1224** when knob **1254** is rotated. Placement in mount **1224** prevents axial movement of catheters **1120**, **1130** with respect to carriage **1220**, **1230**. Each carriage (e.g., **1240**) is defined by a main body portion (e.g., **1242**) that defines a channel (e.g., **1248**) therethrough to at least partially surround rail **2201** of stand **2200**.

(105) With reference to FIGS. **32-33**, when installed in mount **1224**, **1234**, each catheter the proximal gear **1252** that is operably coupled to each catheter meshes with a respective worm gear **1256** that is rotatably situated within each mount and operably coupled to a respective handle **1254**. In this manner, rotation of the handle **1254** causes the worm gear **1256** to rotate about a central axis of rotation. The intermeshing of the winding of the worm gear with the teeth of the gear **1252** causes gear **1252** to rotate about a central axis of catheters **1120**, **1130**. Since gear **1252** is directly or indirectly coupled to the proximal end of each catheter **1120**, **1130**, the catheter **1120**, **1130** also rotates when its respective handle or knob **1254** is rotated. This provides a fine adjustment to help steer each catheter through tortuous anatomy.

(106) As such, each of catheters **1120**, **1130** can be displaced rotationally about a central axis of the system (e.g., about tether **1140**), and with respect to each other. It will be appreciated that any of catheters **1120**, **1130** (or additional catheters combined with catheters **1120**, **1130**) can be utilized with stand **1200** alone, or in combination with other system components. Thus, a double catheter assembly may be used as depicted in FIG. **31**, or a single or double catheter assembly may be used if all three catheters are not required. Catheters **1120**, **1130** can be of any desired length. In accordance with some implementations, catheter **1120** can be between about 90 cm and about 140 cm in length, or any increment therebetween of about one centimeter. In accordance with still further aspects, catheter **1130** can be between about 100 cm and about 160 cm in length, or any increment therebetween of about one centimeter. Tether **1140** can be any desired length, such as between about 120 and 300 cm in length, or any increment therebetween of about one centimeter.

Other than the actuator portion(s) catheters **1120**, **1130** can be the same or substantially the same as catheters **120**, **130**, for example.

(107) As further depicted in FIGS. **31-32**, the actuators **1126**, **1136** can be selectively translated proximally or distally with respect to rail **1201** by unclamping the lock that holds each actuator in place by rotating lock handle **1225**. When a respective lock handle **1225** is loosened, each actuator **1136**, **1136** can be slid along the rail **1201**. When a desired location within a patient's vasculature is reached, the actuator **1126**, **1136** can then be locked in place by tightening its respective lock handle **1225**. This provides for a "coarse" adjustment of the position of each catheter **1120**, **1130**. If desired, a finer adjustment can be provided between the carriage **1220**, **1230**, **1240** and the rail, for example, by forming a gear rack into or on top of rail **1201** that engages a rotational actuator of each carriage (not shown) with gear teeth that engage the rack to permit the position of the carriage **1220**, **1230**, **1240** to be adjusted by rotating a knob of the rotational actuator. The illustrated carriages are depicted with only having the aforementioned coarse adjustment.

(108) With reference to FIGS. **33-34**, each actuator **1126**, **1136** can include a steering mechanism that utilizes one or more tensioned pull wires that are actuated from within the handle (e.g., **1226A**) of the actuator **1126**. FIG. **34** depicts the catheter **1120** and handle **1226A** after it has been lifted out of mount **1224**, and after removal of one half of the outer housing to view interior components. The handle **1226A** is constructed from an outer housing that includes a distal housing section **1273** coupled to a proximal housing section **1278** by one or more longitudinal bridges that pass radially outwardly past a rotatable pull wire actuator handle **1275** defining threads along its interior surface. The hub **1272** including gear **1252** and flange **1272A-1272E** can be formed from a single molding that is received in the distal housing component **1273**. As illustrated, the flange portion of the hub **1272** includes two distally located disc shaped portions **1272B**, **1272C** separated by a channel **1272** defined by an outwardly facing annular surface that contacts ridge or flange **1229** of actuator **1126**. The hub **1272** further includes a proximal enlarged portion **1272A** having a shape that is slidably received within a channel defined in each half of the housing of handle **1226A** to prevent relative rotation of hub **1272** with respect to the housing **1273**, **1278** of the handle **1226A**. On its proximal face, hub **1272** defines a pair of sockets to slidably receive a distal end of one or more rails **1274**. The proximal end of the one or more rails **1274** are received by a proximal collar **1277**. The proximal collar **1277** is fixedly coupled to the proximal end **1122** of the tubular portion of the catheter **1120**. A carriage **1276** defining a plurality of threads thereon is threadably received by the interior threaded surface of handle **1275** and is slidably received over the rails **1274**. Carriage **1276** is attached to a proximal end **1121P** of a pull wire **1121** that is operably coupled to a distal end of catheter **1120**. Movement of the carriage proximally or distally over rails **1274** is effectuated by rotating handle **1275** with respect to the housing of handle **1226A**. The interior threaded surface of handle **1275** pushes the carriage **1276** along the rails. Movement of the carriage **1276** proximally applies tension to the pull wire **1121**, causing the distal tip of the catheter **1120** to deflect off of the catheter's longitudinal axis. Rotation of the handle **1226A** with respect to and within the carrier **1220** permits rotation of the deflected end of the catheter, permitting the distal end of the catheter to be advanced through tortuous anatomy.

(109) FIG. **34** further illustrates that the proximal collar **1277** is held in a fixed rotational and axial position within the housing of the handle **1226A** such that the proximal end **1122** of catheter is also held axially and rotationally in place with respect to the housing of the handle **1226A**. A proximal portion of the handle **1226A** is defined and surrounded by proximal housing portion **1278**, which receives flush line **1262** and inflation line **1264**, discussed below; as well as a proximal cap **1279** and a further lock collar **1269** containing a further circumferential fluid seal to prevent liquids accumulating within handle **1226A** from leaking out.

(110) As further illustrated in FIGS. **31**, **34**, and **35** actuator **1126** includes a valved flush line **1262** in fluid communication with an interior lumen of catheter **1120** and a second lumen **1264** to receive a pressurized fluid to inflate a toroidally shaped inflatable member, or balloon **1265** (FIG. **35**) that

seals against the outer surface of catheter **1130**, to prevent leakage of fluids from a patient past the proximal end of catheter **1120**. FIG. 35 depicts a slight variation **1226A** of handle **1226A** (also illustrated in FIGS. 36 and FIGS. 38-40) showing interior components of the handle **1226**.

Although not visible in FIG. 34, FIG. 35 illustrates additional internal components to which the flush line **1262** and inflation line **1264** attach. Specifically, an interior fluid manifold is depicted that includes a flush housing **1261**, an inflation housing **1263**, an inflatable member **1265** and a proximal lock collar **1269**. The fluid manifold can be formed from components that are integral or separate. Flush line **1262** is in fluid communication with an interior volume (not shown) within flush housing **1261** that in turn is in fluid communication with the interior lumen of catheter **1120**. In this manner, a flushing fluid, or other beneficial agent, can be introduced to a tissue region being treated by directing the fluid through line **1262**, into the chamber within flush housing **1261**, and down the lumen of catheter **1120** to the distal end of catheter **1120** where the fluid is ejected at the location proximate tissue being cut during an electrosurgical process, for example. The beneficial agent or flush can include a dextrose flush, for example, to the area being treated.

(111) With continuing reference to FIG. 35, adjacent to and proximal to the flush housing **1261** is an inflation housing **1263** having an interior cavity (not shown) that is in fluid communication with inflation line **1264**. The chamber within inflation housing **1263** is in turn in fluid communication with annularly or toroidally shaped inflatable member **1265**, which is defined by inner and outer flexible surfaces that deflect when the inflatable member **1265** is inflated by directing pressurized fluid through inflation line **1264**. The annular inflatable member inflates radially inwardly to seal around the outer surface of catheter **1130**, and radially outwardly to occupy volume within the housing of the handle **1226A** to prevent leakage of flushing fluid, for example, out of handle **1226A**, or bodily fluids from a patient. The fluid manifold terminates at a proximal end with lock collar **1269**, which can comprise an internally threaded outer member that can rotate with respect to an externally threaded inner member and a compressible seal between the two components to cause the compressible seal to be urged against the outer surface of catheter **1130** when the lock collar **1269** is tightened.

(112) A detailed illustration of tensionable anchor **1146** is depicted in FIG. 36. Anchor **1146** includes a housing having a slidable handle **1145** that slides along a slot **1148** (FIG. 31, FIG. 37) in the outer surface of the housing of the anchor **1146**. Handle **1145** is operably coupled to a proximal region, or a proximal end, of tether **1140** in any of a variety of manners. In the illustrated implementation, the handle **1145** can be coupled to a shaft (not shown) that passes through a shuttle **1141** that in turn is connected to a coil spring **1143**, wherein the coil spring **1143** is coupled to an interior portion of the housing of anchor **1146**. The tension on the coil spring **1143** can be adjusted by rotating handle **1145**, which then applies a proximally directed force on a proximal end or end region of the tether **1140** to apply tension to the tether **1140** when the distal anchor coupled to the tether is anchored in tissue. FIG. 36 depicts a half of the housing of anchor **1146** as being removed, and presented in FIG. 37. The shuttle **1141** includes one or more bosses that ride in a channel **1242A** formed into the housing of the anchor **1146**. An anchor block **1141A** (FIG. 37) is operably coupled to the proximal end of the tether **1140**. Anchor block **1141** can be slidably received (e.g., along a vertical direction) in shuttle **1141**.

(113) FIG. 35 depicts a slight variation of the actuator assembly **1100** showing three actuators **1116'**, **1126'**, **1136'** having components and functionality similar to actuator **1126**, **1136** that are configured to be coupled to three concentric catheters, such as catheters **110**, **120**, **130**. A further anchor **1146'** is depicted that applies tension to a central tether, wherein the anchor includes an arm that is attached to the tether, wherein the arm pivots about a pivot point, and is spring loaded to apply tension to the tether. The actuators are slidably mounted to a rail **1201** that in turn is coupled to a base frame **1200** that is clamped to a stool or other object by a clamp **1200A**. FIGS. 39A and 39B illustrate vies of the manner in which the actuator assembly can be coupled to a stool or other object. Specifically, the rail **1201** is coupled to a base portion **1200** that rests on a surface, such as a

table or stool, that can be clamped in place using a clamp **1200A** that wraps around an edge of the surface on which the base portion is sitting. In the illustrated implementation, the clamp is actuated by up and down movement. If desired, the rail **1201** may include measurement indicia along its surface and if desired may be provided with additional stoppers (not shown) that can be received over the rail **1201**, slid to a desired position, and tightened in place against the rail **1201**. The stoppers can be placed in locations on either side of one or more of the actuators to limit the degree of travel of the respective actuator along the rail **1201**.

(114) In further accordance with the disclosure, a variety of methods are provided of performing a medical procedure using the actuator assembly and catheter arrangements described herein. For example, a system of nested catheters (e.g., **110**, **120**, **130**) can be directed or advanced to a target location inside a patient. A distal end of an outermost tubular catheter (e.g., **110**) can be placed in a first location. A distal end of a further tubular catheter (e.g., **120**) can be placed in a second location located distally with respect to the first location. A distal end of an inner tubular catheter (e.g., **130**) can be placed in a third location located distally with respect to the second location. The method can then include performing any one or more of a variety of a therapeutic or diagnostic procedures using at least one of the catheters (e.g., **110**, **120**, **130**).

(115) In some implementations, a distal region of the outermost catheter (e.g., **110**) can be disposed in an aortic arch of a patient, a distal region of the further tubular catheter (e.g., **120**) can be disposed through a cardiac valve of a patient, and the innermost tubular catheter (e.g., **130**) can be manipulated to perform a therapeutic or diagnostic procedure, such as a SESAME procedure as set forth herein, or other procedure(s). As such, the therapeutic procedure can include, in some implementations, cutting into a left ventricular outflow tract obstruction to increase the effective cross-sectional area of the left ventricular outflow tract.

(116) In some implementations, the innermost tubular catheter can include a microcatheter, and the method may include directing the microcatheter into a right ventricle of a patient, through a septum of the patient into the left ventricle of the patient, and cutting into a left ventricular outflow tract obstruction to increase the effective cross-sectional area of the left ventricular outflow tract, or other procedure. In some implementations, the first tubular catheter can include a delivery catheter to deliver a beneficial agent (e.g., liquid medicament, saline, dextrose solution, and the like) or a medical device (e.g., stent, shunt, or the like) to an anatomical location. One or more of the catheters can include one or more visualization markers (e.g., viewable under fluoroscopy or MRI) and/or one or more electrodes or one or more cutting elements to cut through anatomical tissue. The diagnostic or therapeutic procedure can be selected from the group consisting of a MIRTH procedure, a LAMPOON procedure, a PASTA (Pledget-assisted suture tricuspid valve annuloplasty) procedure, an ELASTIC procedure, a BASILICA procedure, a robotic surgical procedure, a cerclage procedure, or delivery of a medical device. In one illustration of FIG. **39C**, the system of the disclosure is set up to perform a MIRTH procedure.

(117) A MIRTH (Myocardial Intramural Remodeling by Transvenous Tether) procedure is a transcatheter ventricular remodeling procedure, wherein a transvenous tension element is placed within the walls of the beating left ventricle and shortened to narrow chamber dimensions. Further details of the MIRTH procedure can be found in U.S. patent application Ser. No. 18/151,601, which is incorporated by reference in its entirety for all purposes. The MIRTH method can be used to reduce the dimensional size of a portion of a patient's heart. The method can include advancing a guidewire into a patient's circulatory system and into the patient's heart, advancing the guidewire through the myocardium (between layers of cardiac tissue around a chamber of a heart, for example) to define a passageway around at least a portion of the heart between an outer surface of the heart and an inner surface of the heart, exchanging the guidewire with an implant including a tensioning element so that the tensioning element traverses the passageway, advancing a lock over the tensioning element, applying tension to the tensioning element to change the dimensional size of a portion of the heart, and locking the lock to maintain the tension in the tensioning element.

Such a technique can be performed utilizing system **100**, or components thereof. Various supporting catheters are useful in performing a MIRTH procedure. In the context of the present disclosure, a two-catheter system can be used, as illustrated in FIG. **39C**, wherein catheters **110** and **120**, for example, are used to steer to the myocardium and perform a MIRTH procedure. A further illustrative procedure is set forth in International Application No. PCT/US2020/045674, which is incorporated herein by reference, wherein the disclosed catheter system **100** can be used to perform this procedure.

(118) FIGS. **40A-40D** depict further implementations of a catheter system in accordance with the present disclosure, or aspects thereof, having a different actuator assembly than the system of FIGS. **31-40**. For purposes of reference, the proximal and distal directions of the assembly are indicated in various figures. This illustrated implementation of the actuator includes a variation with respect to that of FIGS. **31-39C**, wherein the closing of the cover **2234A** pushes downwardly on a pin **2280** that rotates a gear **2288** that pushes an inner support or journal **2290** upward that rotatably receives the worm gear **2256**. The journal **2290** is slidably disposed within the housing of the actuator assembly, and is biased into a downward location by a tension spring **2292**, wherein, when the cover **2234A** is open, as illustrated in FIG. **40C**, the spring **2292** pulls the journal or support **2290** downward, causing the worm gear **2256** to lower and disengage from gear **2252**, permitting the gear **2252**, and thus the catheter seated in the actuator, to rotate about its axis without mechanical interference from the worm gear **2256**. When a desired rotational orientation is obtained for the catheter with the cover **2234A** open, the cover can then be closed, causing a boss **2234B** formed on the underside of the cover to push downwardly on an upper end surface **2282** (FIG. **40B**) of pin **2280**. With reference to FIG. **40C**, downward motion of pin **2280** causes a gear rack **2284** formed into a lower surface of the pin **2280** to cause rotation of a sprocket or gear **2288** about an axle **2286**. The teeth of gear **2288** are further engaged with a complementary toothed rack on journal **2290**, causing journal **2290** to be pushed up against the force of the tension spring **2292**, resulting in mechanical engagement between the worm gear **2256** and the gear **2252** that surrounds and is operably coupled to a respective catheter (e.g., **130**).

(119) With reference to FIG. **40A**, the actuator assembly can be similar in design and operation to those illustrated in FIGS. **31-39C**, with the addition of the above-described mechanism, which permits a coarse rotational adjustment of a catheter about its longitudinal axis when open. Thus, like reference numerals for the rail **2201** and other components are meant to parallel and mirror the embodiment of FIGS. **31-39C**.

(120) The top view of the assembly in FIG. **40B** illustrates the cover **2234A** in an open position, showing the relative location of boss **2234B** that contacts and pushes down on the upper surface **2282** of pin **2280**. A similar dial adjustment **2275** can be operably coupled to a pull wire as with the embodiment of FIGS. **31-39C** and the same flush and sealing ports can be provided.

(121) As mentioned above, with reference to FIG. **40C**, when the top cover **2234A** is open, the tension spring **2292** goes back to its contracted rest position and pulls the worm gear **2256** and its journal or mount **2290** downwardly so the worm gear **2256** disengages from the gear **2252**. At this time, the catheter is free to rotate within the housing of the carriage.

(122) As illustrated in FIG. **40D**, when the carriage top cover **2234A** is closed, the boss **2234B** inside the top cover **2234A** pushes the pin **2280** down. The pin rotates the gear **2288** and the spring **2292** stretches and the worm gear **2256** goes up to engage gear **2252** such that the catheter rotational position is locked. Under this condition, spring **2292** is actually under tension, although this is not directly illustrated in FIG. **40D**.

(123) In further accordance with the disclosure, FIG. **41** depicts a side view of a distal portion of a further catheter **1330** in accordance with the present disclosure configured to deploy a tissue anchor out through a lateral side port of the catheter. As illustrated in FIG. **41**, catheter **1330** is formed from an outer tubular member **1335** that contains a deployable anchor that can be deployed through a side port **1340** defined through tubular member **1335**. Catheter further includes an electrode **1337**

that includes an electrically exposed portion (not shown) to permit the completion of an electrical circuit through tissue being cut during an electrosurgical procedure. The electrically exposed portion can exist along the outer edge of the arch facing outwardly, wherein insulating material remains around the remaining circumference of the electrode **1337**. The electrode **1337**, as depicted, extends through a respective side port **1337A** defined through wall **1335** of the catheter **1330**. But, it will be appreciated that the electrode can simply be formed into or protruding from the sidewall without a side port **1337A** as with electrode **137** illustrated in other embodiments herein. The electrode **1337** can be configured to be deployed outwardly from the side port **1337A**, for example, by pushing the electrode to apply a column force or axial compression to the electrode, causing the electrode **1337** to protrude from the port **1337A**. Similarly, the electrode can be spring loaded or otherwise be biased to extend outwardly through the port **1337A**, wherein the electrode can be pushed laterally back into the port by withdrawing the catheter **1330** proximally into a guiding catheter or sheath, for example. As illustrated, catheter **1330** further includes a distal marker band **1360** which can comprise a band of radiopaque material or be configured to be visible under additional or alternative visualization modalities as set forth herein.

(124) FIGS. **42A-42D** are images of the catheter of FIG. **41** illustrating successive stages of deploying tissue anchor **1350** out through the lateral side port **1340** of the catheter **1330**. FIG. **42A** illustrates catheter **1330** prior to deployment of the anchor **1350**, wherein the relative locations of the anchor exit port **1340**, distal marker band **1360**, electrode **1337**, and side electrode port **1337A** are presented. FIG. **42B** illustrates an initial step of deployment of anchor **1350**, wherein a pair of tines **1354** begin exiting out of port **1340**, which is located distally with respect to the electrode **1337**. In other implementations, the lateral anchor port **1340** can be located proximally with respect to the electrode **1337**. FIG. **42C** illustrates the tines **1354** of the anchor when they are about halfway deployed, and FIG. **42D** illustrates the tines **1354** of the anchor **1350** when fully deployed. As illustrated in FIGS. **43-44**, the anchor **1350** is comprised of an elongate wire **1356** that is slidably disposed along a lumen defined along the interior of catheter **1330** inside tubular body **1335** that is in fluid communication with the exit port **1340**. Pushing the wire **1356** distally with respect to the outer tubular body **1335** causes the anchor **1350** to deploy, wherein the tines **1354** of the anchor **1350** are directed proximally to facilitate the anchoring of the tines **1354** of the anchor **1350** into tissue. The tines **1354** can be coupled to the wire **1356** at a junction **1352**, wherein the junction can prevent rotation or flexing between the wire and tines **1354**, or the junction **1352** can be a swiveled coupling, if desired. As will be appreciated, the two-tined design of anchor **1350** facilitates lateral deployment, whereas the three-tined design of implementation **150** (FIG. **14**) is well suited for deployment out of a distal passage of a catheter (e.g., **130**).

(125) FIGS. **45A-45B** illustrate aspects of utilizing a catheter-based system (e.g., **100**) in accordance with the present disclosure to perform a Laceration of the Anterior Mitral leaflet to Prevent Outflow Obstruction (LAMPOON) procedure, which is a transcatheter electrosurgical technique to split the anterior mitral valve leaflet immediately prior to transcatheter mitral valve replacement (TMVR). As illustrated, to facilitate this procedure, catheter **110** is deflected over the aortic arch into the ascending aorta, wherein the distal end of the catheter **110** is outside of the aortic valve. Catheter **120** is deployed distally of catheter **110** until the distal end of catheter **120** is just under the aortic valve. Catheter **130**, which include a cutting element **137**, is deployed and steered to the wall of the left ventricle where anchor **150** is deployed out of the distal end of catheter **130** and into the wall of the left ventricle. Catheter **130** can then be retracted proximally to reveal a tether in the form of a hypotube **140'** that has sufficient column strength to help insert anchor **150** into the tissue. The catheter **130** can then be aligned such that the cutting element **137**, whether electrosurgical, a sharp surface, or an ultrasonic cutter is placed proximate the anterior mitral leaflet to cut through the anterior mitral leaflet by starting cutting at a bottom edge of the anterior mitral leaflet, and then reciprocating back and forth to cut through the leaflet. The catheter **130** can be selectively withdrawn to maintain the cutting element in an appropriate position to

continue cutting the leaflet until the cutter **137** is near the anterior mitral annulus, and the anterior mitral leaflet has been split. After splitting the valve leaflet, the replacement valve can be installed. The rigidity of element **140'**, whether provided as a solid element or a tubular element cooperates with the stiffness of the catheter **130** to help position the cutter in a correct location to continue cutting tissue.

(126) FIGS. **46A-46B** illustrate aspects of utilizing a catheter-based system in accordance with the present disclosure to perform a BASILICA procedure for cutting a leaflet of the aortic valve. In this implementation, catheter **120** deflects over the aortic arch and down into the ascending aorta. Catheter **130** supports a crossing wire **370** inside a cusp of the aortic valve. The crossing wire **370** is advanced through the aortic valve leaflet near a radially outer edge of the leaflet, such as by electrifying the wire **370** wherein the wire **370** includes an electrically exposed distal end and includes an electrically insulating jacket proximal to the distal end of the wire **370**. A snare catheter **310** is introduced alongside catheters **120**, **130** including a tubular member with a snare **312** terminating in a loop that surrounds and can capture the distal end of the wire **370** after the wire **370** has formed a hole through the aortic valve leaflet. The wire **370** is pulled into the distal end of catheter **310**. At this point in the procedure, the wire **370**, pulled into and held in place in catheter **310** can then provide a rail or guide that can permit a cutter **137** of any suitable form to enlarge the hole formed by wire **370**. The wire **370** can be pulled further into catheter **310** to apply tension to wire **370** to provide a lateral force component to catheter **130**, and cutter **137** to cause the cutter to cut through the aortic valve leaflet. This procedure can be repeated on the other two aortic valve leaflets.

(127) FIGS. **47A-47B** illustrate aspects of utilizing a catheter-based system in accordance with the present disclosure to perform an ELASTIC (Electrosurgical Laceration of Alfieri STItCh (ELASTIC) procedure to remove a mitral clip or Alfieri stitch to facilitate placement of a TMVR. In this procedure, catheter **120** is deployed over the atrial septum from right to left toward the mitral valve and left ventricular apex. Catheter **130** is deployed distally to the apex of the left ventricle and anchor **150** is deployed in the tissue in that location. The catheter **130** is retracted proximally until the cutter **137** is in the region of the Alfieri stitch, or the mitral clip attaching the anterior and posterior mitral leaflets to each other. The cutter **137** is aligned with tissue near the stitch, and the cut is made through tissue to separate the two leaflets, permitting installation of a TMVR. The tether **140'** can include a rod or tube to provide sufficient stiffness to help guide the cutting in cooperation with the movement of catheter **130** bearing cutter **137**.

(128) By way of further example, instead of being configured for manual operation, the actuator assembly can be configured to be received by an actuator hub of a robotic surgical system. The system (e.g., **100**) can be introduced into a patient by way of a femoral access point, carotid, or jugular access point, depending on the procedure being performed. The system can be sized and configured to be introduced to a patient's vasculature to perform a procedure within the neurovascular region of a patient. It will be appreciated that system (e.g., **100**) can be used in a variety of percutaneous cardiovascular, thoracic, or laparoscopic procedures, as well as accessing the brain of a patient via the circulatory system or sinus passages of a patient. The systems of the disclosure can be further used to access the digestive system of a patient as well as the urinary or reproductive system of a patient.

(129) The disclosure also provides an electrosurgical system including a radio frequency power supply (such as that described in U.S. Pat. No. 6,296,636, which is incorporated by reference herein in its entirety for any purpose whatsoever) operably coupled to the electrode(s) (e.g., **137**, **137**) and anchor, if desired (e.g., **150**). The radio frequency power supply can be operably (and selectively) coupled to the electrodes by way of a cable or other conductors. Any suitable power level and duty cycle can be used in accordance with the disclosed embodiments. For example, continuous duty cycle (cutting) radiofrequency ("RF") energy can be used, for example, at a power level between about 10 and about 50 Watts, for example, or any increment therebetween of about one watt. The

cuts can be made by applying power for between about one half of a second and about five seconds, or any increment therebetween of about one tenth of a second. The electrosurgery generator can be the Medtronic Force FX C Generator that achieves 5 W to 300 Watts of monopolar radiofrequency (RF) energy, for example.

(130) Implementations of the guidewires for use with the disclosed embodiments preferably include a sterile, single use device intended to cut soft tissue. References to dimensions and other specific information herein is intended to be illustrative and non-limiting. In one implementation, the disclosed guidewire has an outer diameter of 0.035" and a working length of 260-300 cm. The proximal end of the disclosed guidewire, which has no patient contact, can be un-insulated to allow for connection to an electrosurgery generator if desired.

(131) The devices and methods disclosed herein can be used for other procedures in an as-is condition, or can be modified as needed to suit the particular procedure. This procedure for cutting the myocardium can be used in support of a variety of procedures. Likewise, while it can be appreciated that a monopolar cutting system is disclosed, in certain implementations, it is also possible to configure the system to operate in a bipolar configuration. During the step of myocardium laceration, the system can be configured to deliver energy to the myocardium with electrosurgical pads coupled to the patient to complete the circuit. When lacerating the myocardium or other structure with the bent denuded cutting wire, most of the energy is still dissipated in the patient.

(132) In view of the many possible embodiments to which the principles of this disclosure may be applied, it should be recognized that the illustrated embodiments are only preferred examples of the disclosure and should not be taken as limiting the scope of the disclosure. Each and every patent and patent application referenced herein is expressly incorporated by reference herein in its entirety for any purpose whatsoever.

Claims

1. A method of performing a medical procedure, comprising: providing a medical device, the medical device including: an elongate tether having a proximal end configured to be anchored to a proximal anchor point outside of a patient's anatomy and a distal end having a tissue anchor disposed thereon to penetrate through myocardial tissue to permit tension to be applied to and maintained along the elongate tether when the proximal end of the elongate tether is anchored to the proximal anchor point and the tissue anchor is anchored into myocardial tissue; a first tubular catheter surrounding the elongate tether and being slidably displaceable along and with respect to the elongate tether when the elongate tether is under tension, the first tubular catheter having a proximal end, a distal end, and a cutter; and a second tubular catheter surrounding the first tubular catheter and being slidably displaceable along and with respect to the first tubular catheter, the second tubular catheter having a proximal end and a distal end; directing the medical device to a target location inside a patient's left ventricle proximate a native tissue obstruction in a left ventricular outflow tract of the patient; placing the distal end of the second tubular catheter in a first location located proximally with respect to the tissue obstruction; placing the distal end of the first tubular catheter in a second location located distally with respect to the tissue obstruction; penetrating the myocardial tissue with the tissue anchor to secure the distal end of the elongate tether in place with respect to the myocardium; orienting the cutter of the first tubular catheter toward the native tissue obstruction; and reciprocating the first tubular catheter along the elongate tether to cause the cutter to cut into the native tissue.

2. The method of claim 1, wherein the first tubular catheter includes a distally-facing flush port configured to direct a pressurized fluid over the cutter of the first tubular catheter, and wherein the method further includes directing the pressurized fluid out of the distally-facing flush port over the cutter while reciprocating the first tubular catheter.

3. The method of claim 1, wherein the cutter includes an electrode operably coupled to electrical circuitry that is configured to determine a state of the anatomical tissue.
 4. The method of claim 1, wherein the cutter includes an ultrasonically-driven cutter.
 5. The method of claim 1, wherein the cutter is operably coupled to electrical circuitry, wherein the electrical circuitry is configured to correlate a voltage drop or a current drop of the cutter with a state selected from the group consisting of (i) a state of tissue being cut by the electrode, (ii) a state of fouling of the electrode.
 6. The method of claim 2, wherein the pressurized fluid is a dielectric fluid.
 7. The method of claim 6, wherein the dielectric fluid includes a dextrose solution.
 8. The method device of claim 1, wherein the medical device further includes an actuator assembly coupled to a proximal region of the elongate tether, a proximal region of the first tubular catheter, and a proximal region of the second tubular catheter.
 9. The method device of claim 8, wherein the medical device further includes a third tubular catheter surrounding the second tubular catheter and being slidably displaceable along and with respect to the second tubular catheter, the third tubular catheter having a proximal end and a distal end, and further wherein a proximal end of the third tubular catheter is coupled to the actuator assembly.
 10. The method of claim 9, wherein the actuator assembly further includes a first actuator operably coupled to a proximal end of the first tubular catheter, and wherein the first actuator is configured to advance and retract the first tubular catheter proximally and distally over the elongate tether when the elongate tether is under tension.
 11. The method of claim 10, wherein the actuator assembly further includes a second actuator operably coupled to a proximal end of the second tubular catheter, wherein the second actuator is configured to advance and retract the second tubular catheter proximally and distally over the first tubular catheter, and further wherein the distal end of the second tubular catheter is capable of being retracted proximally past the distal end of the first tubular catheter.
 12. The method of claim 11, wherein the actuator assembly further includes a third actuator operably coupled to a proximal end of the third tubular catheter, wherein the third actuator is configured to advance and retract the third tubular catheter proximally and distally over the second tubular catheter, and further wherein the distal end of the third tubular catheter is capable of being retracted proximally past the distal end of the first tubular catheter and the distal end of the second tubular catheter.
 13. The method of claim 1, wherein each of the first tubular catheter and second tubular catheter to permit an operator to determine relative axial and rotational location of each of the first tubular catheter and second tubular catheter relative to each other and surrounding anatomy.
 14. The method of claim 1, further comprising conforming a shape of the first tubular catheter to match that of the native tissue obstruction such that the first tubular catheter is in at least substantially continuous contact with the native tissue obstruction along a length of the tissue obstruction.
 15. The method of claim 1, wherein the first tubular catheter is reciprocated along the elongate tether to cut an incision into the obstruction to enlarge an effective cross-sectional area of the left ventricular outflow tract of the patient.
 16. The method of claim 1, wherein the first tubular catheter is reciprocated along the elongate tether to cut an incision into the obstruction until the cutting element traverses a straight path between the distal end of the second tubular catheter and the tissue anchor.
 17. The method of claim 1, wherein the cutter is formed at least in part from radiopaque material.
 18. The method of claim 1, wherein the cutter includes a diamond coated wire.
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